

Use of GIS in Planning

ARCH 305: Architecture and Town Planning

What is GIS?

- **Geographic Information Systems** (GIS) are computerized systems designed for the storage, retrieval and *analysis* of geographically referenced data

- GIS uses advanced analytical tools to explore at a scientific level the spatial relationships, patterns, and processes of cultural, biological, demographic, economic, geographic, and physical phenomena

Tools for GIS

- Hardware
  - Computer
  - Digitizer
  - Scanner
  - Printer/Plotter
- Software
  - Desktop GIS
  - Internet GIS
  - CAD Software
  - Database Software
- Multimedia (photos, videos, 3D models)

Unique capabilities of GIS

- GIS stores related geographic features in separate collections of files called map layers

- Map layers can be reused easily and assembled into any number of map compositions and overlaid for analysis

GIS answers the following

- Location: What is at...? Where is it?
- Condition: Status of features?
- Trends: What has changed since...?
- Patterns: What spatial patterns exist?
- Modeling: What if…?

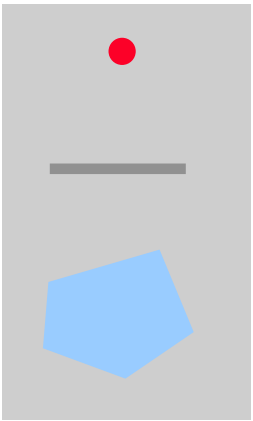
Scale of GIS data

Global to local



Vector data

- **Map features**
  - Points, lines, polygons



- **Feature attributes**
  - Every feature has attributes (e.g. name, area, population)

Shape	Name	Class	Pop2000	State
Point	New York	City	8,008,278	NY
Point	Los Angeles	City	3,694,820	CA
Point	Chicago	City	2,896,016	IL

Raster Data

Stored electronic image or picture taken as an aerial photograph or satellite image



composed of a rectangular array of square cells, called pixels, with a number in each cell representing the solid color fill of that cell

GIS DATA AND LAYERS

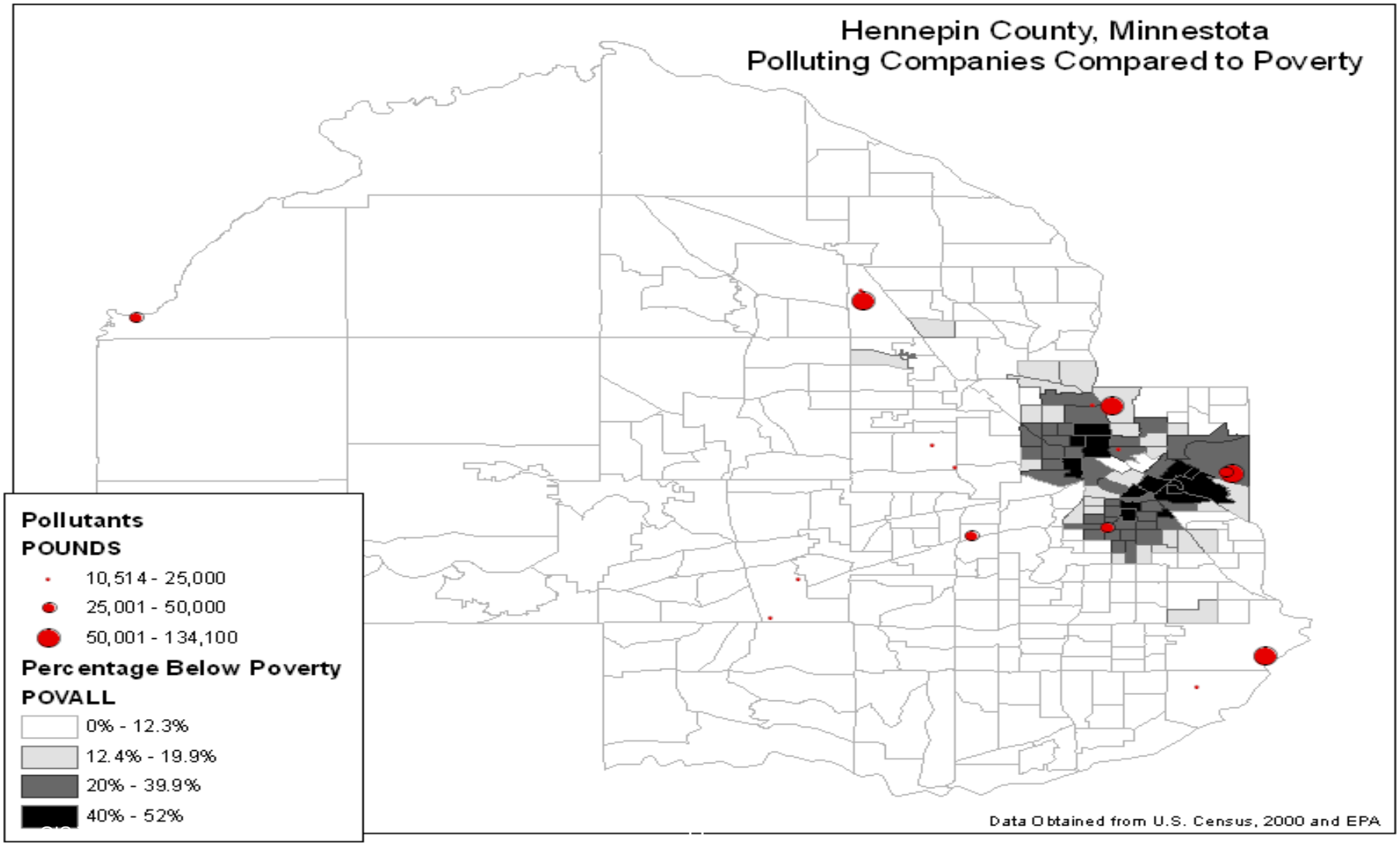
Databases

Not easy to interpret

FID	Shape *	STFID	TRACT	MEDINC	POVALL	POVU18	POVO65	POVFAM
0	Polygon	27053000101	000101	39159	16	28	3	12
1	Polygon	27053000102	000102	36563	13	20	5	10
2	Polygon	27053000300	000300	48664	5	3	7	3
3	Polygon	27053000601	000601	40863	11	22	4	8
4	Polygon	27053000603	000603	50256	2	0	2	1
5	Polygon	27053001100	001100	41099	10	16	13	8
6	Polygon	27053001700	001700	34306	13	16	10	11
7	Polygon	27053002200	002200	28036	41	48	30	31
8	Polygon	27053002400	002400	35029	24	34	19	21

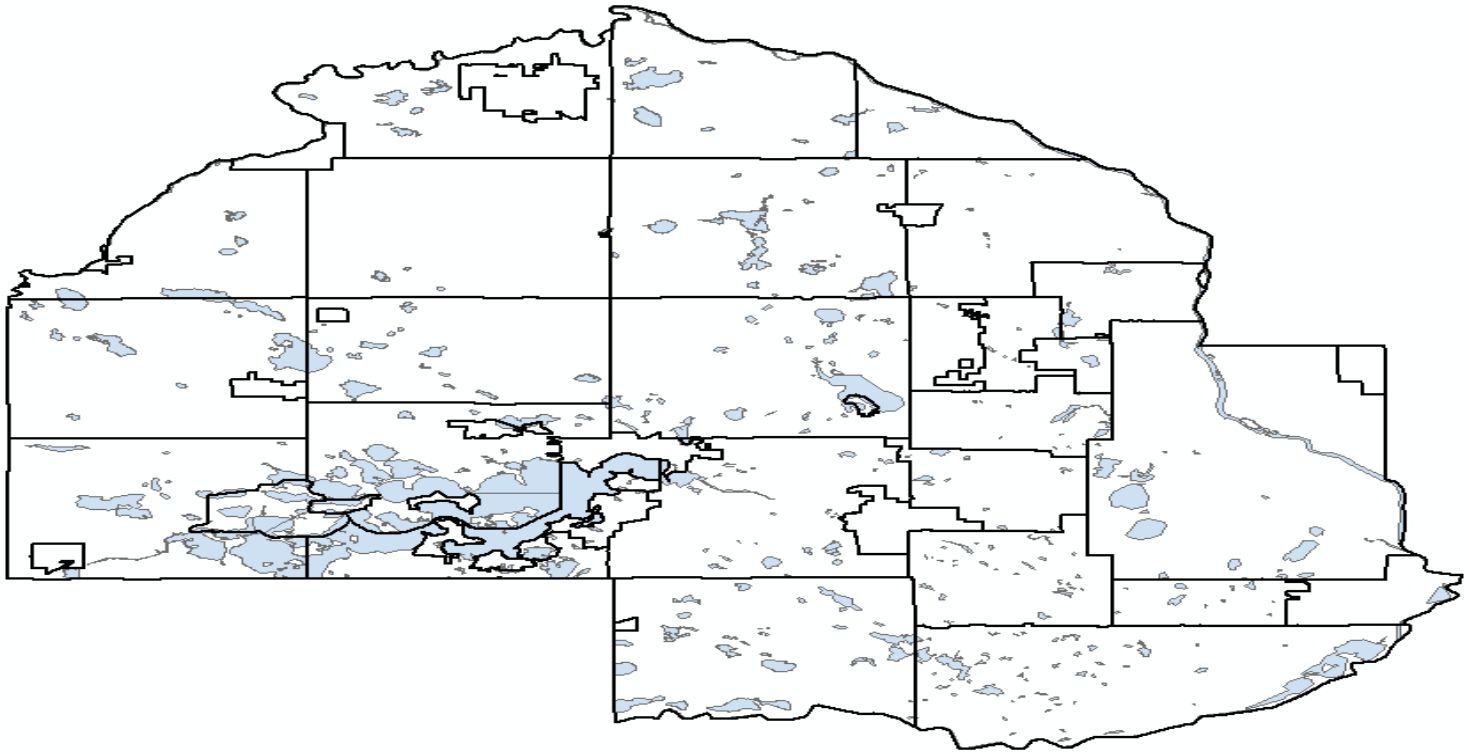
	A	B	C	D	E
1	CITY	SIC_CODE	LATITUDE	LONGITUDE	POUNDS
2	MINNEAPOLIS	26 Paper And Allied Products	44.98244	-93.21470	134100
3	BROOKLYN PARK	30 Rubber And Misc. Plastics Products	45.08788	-93.39455	69380
4	MINNEAPOLIS	49 Electric, Gas, And Sanitary Services	45.02357	-93.27372	66074
5	SAINT PAUL	45 Transportation By Air	44.87071	-93.19908	65000
6	ROCKFORD	30 Rubber And Misc. Plastics Products	45.07797	-93.74738	45304
7	EDEN PRAIRIE	30 Rubber And Misc. Plastics Products	44.87286	93.39926	41600
8	MINNEAPOLIS	34 Fabricated Metal Products	44.94958	-93.27600	34364
9	MINNEAPOLIS	34 Fabricated Metal Products	44.94389	-93.34222	32115
10	MINNEAPOLIS	34 Fabricated Metal Products	44.98259	-93.21810	26140

Data shown as GIS layers



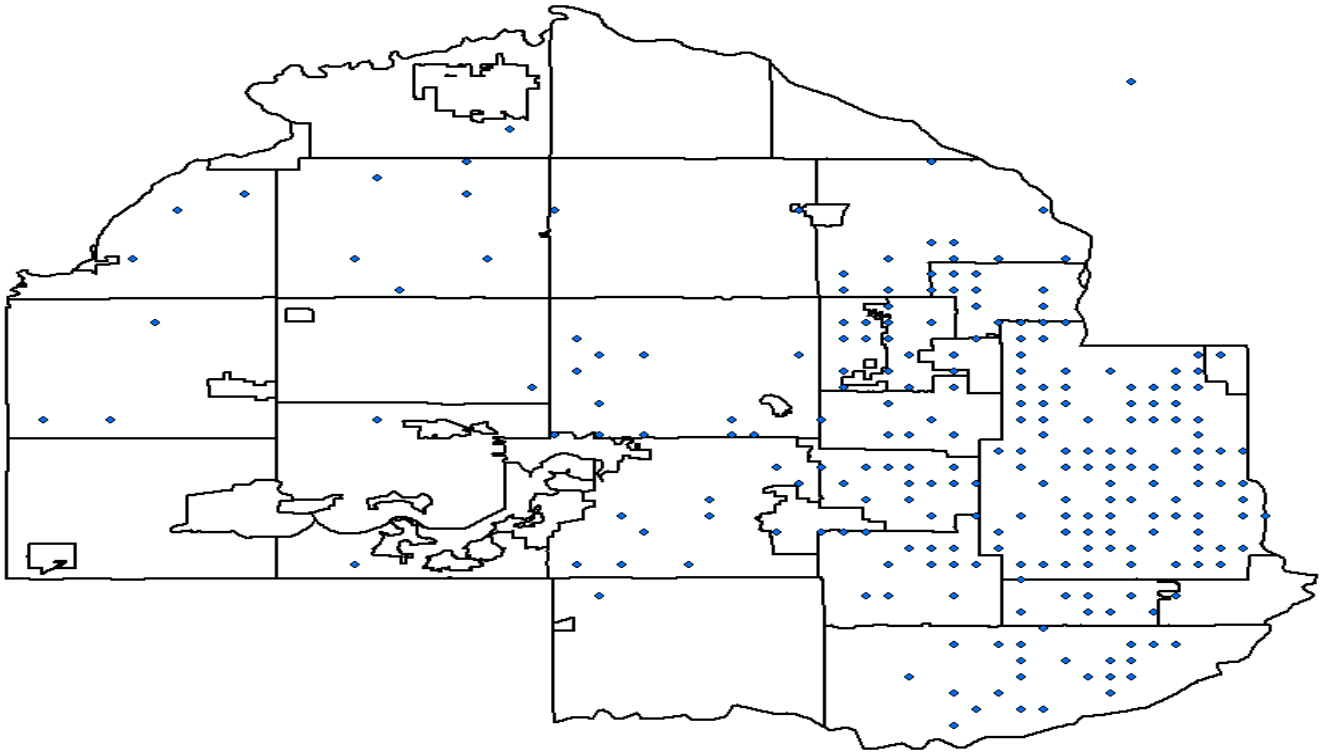
Additional layers

Physical features (lakes, rivers, etc.)



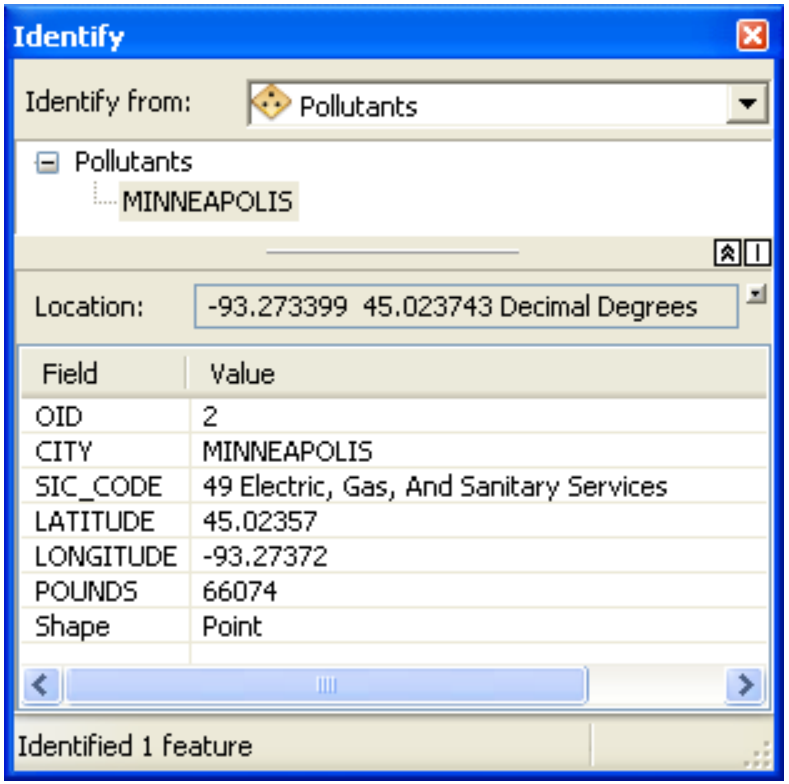
Additional layers

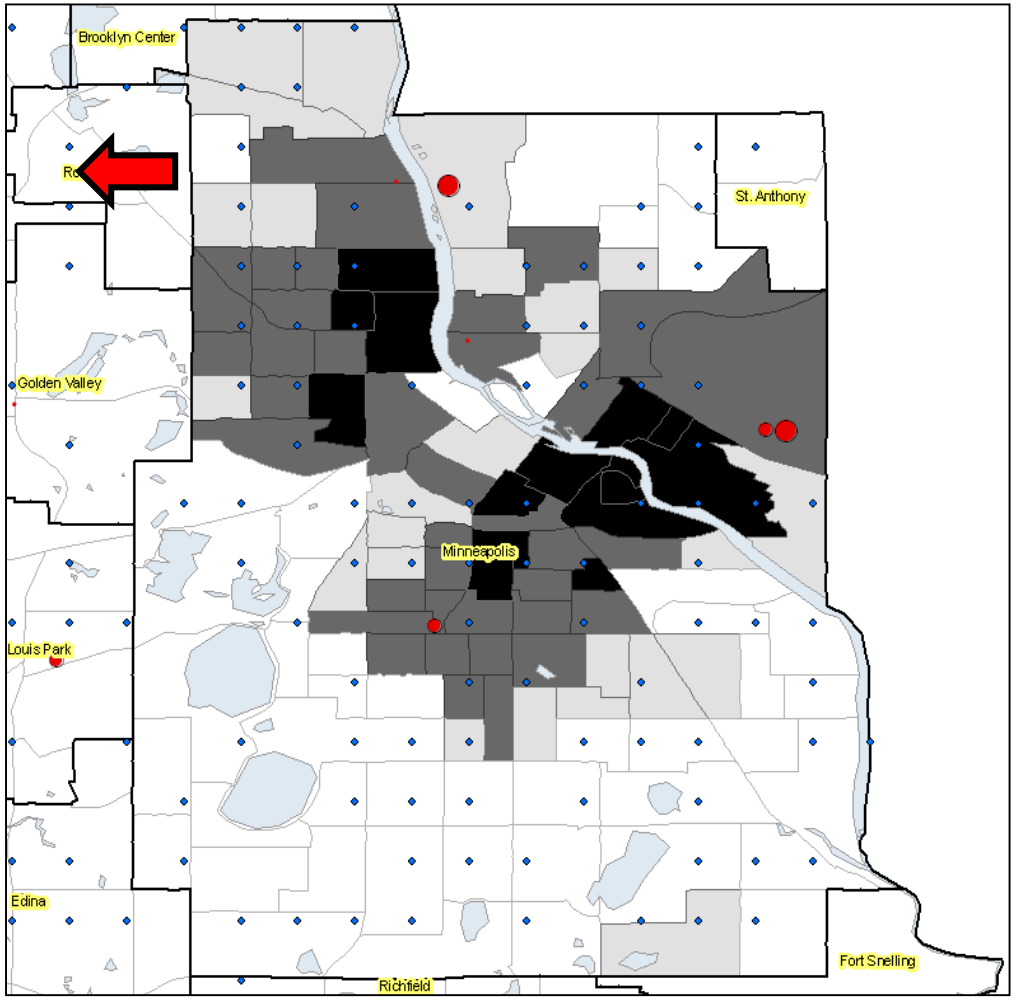
Administrative data (schools)



Maps and tables are interactive

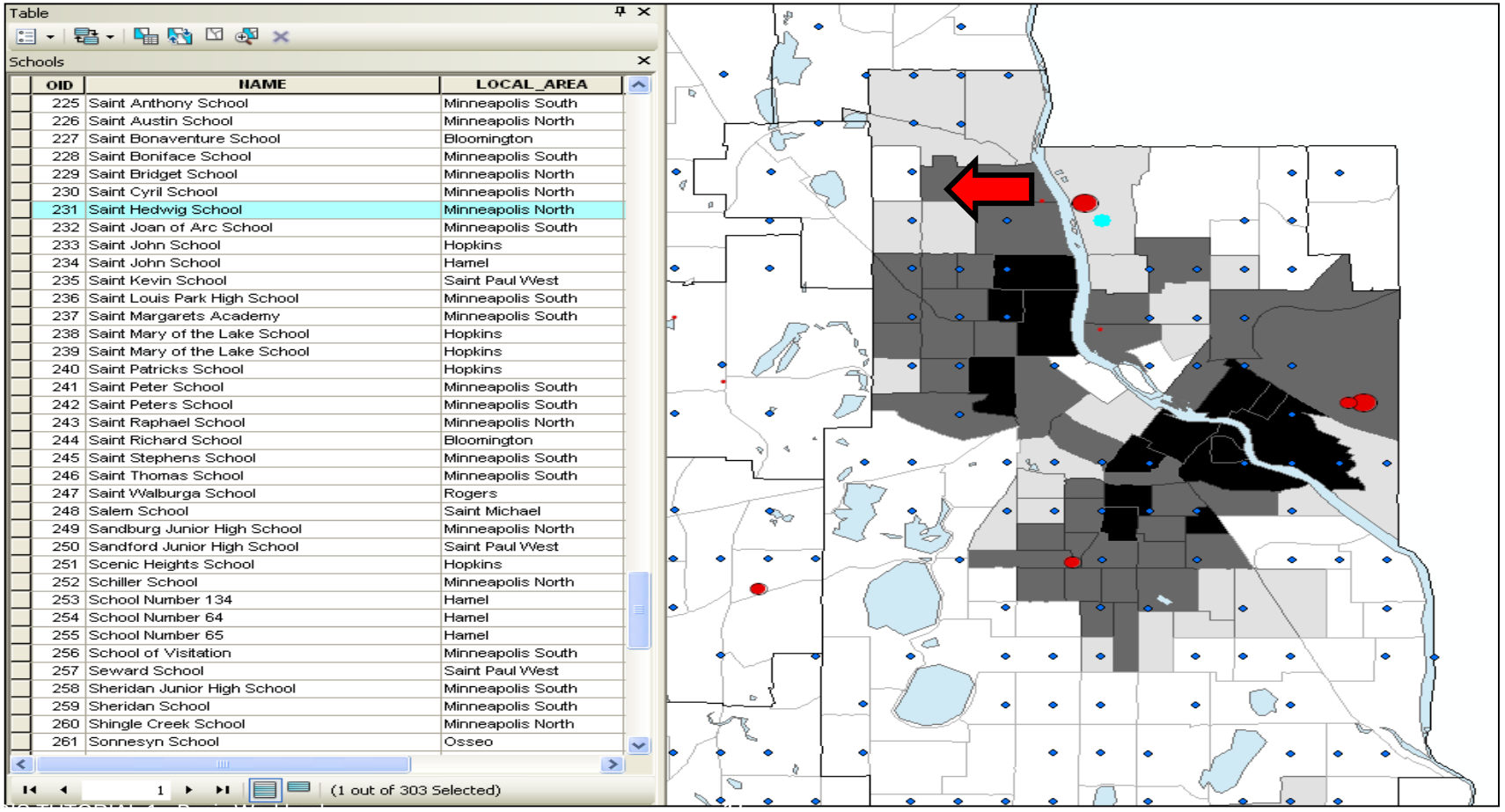
Identify features





Maps and tables are interactive

Select features

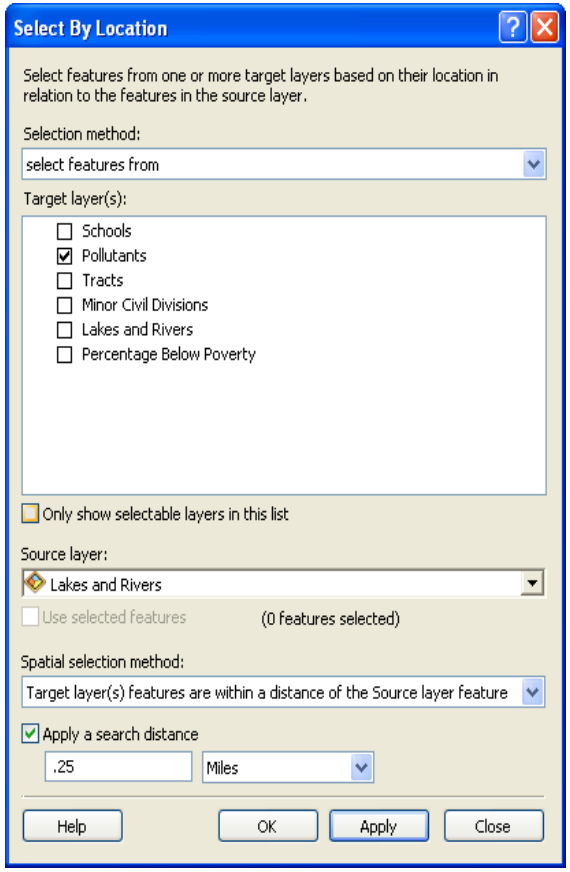


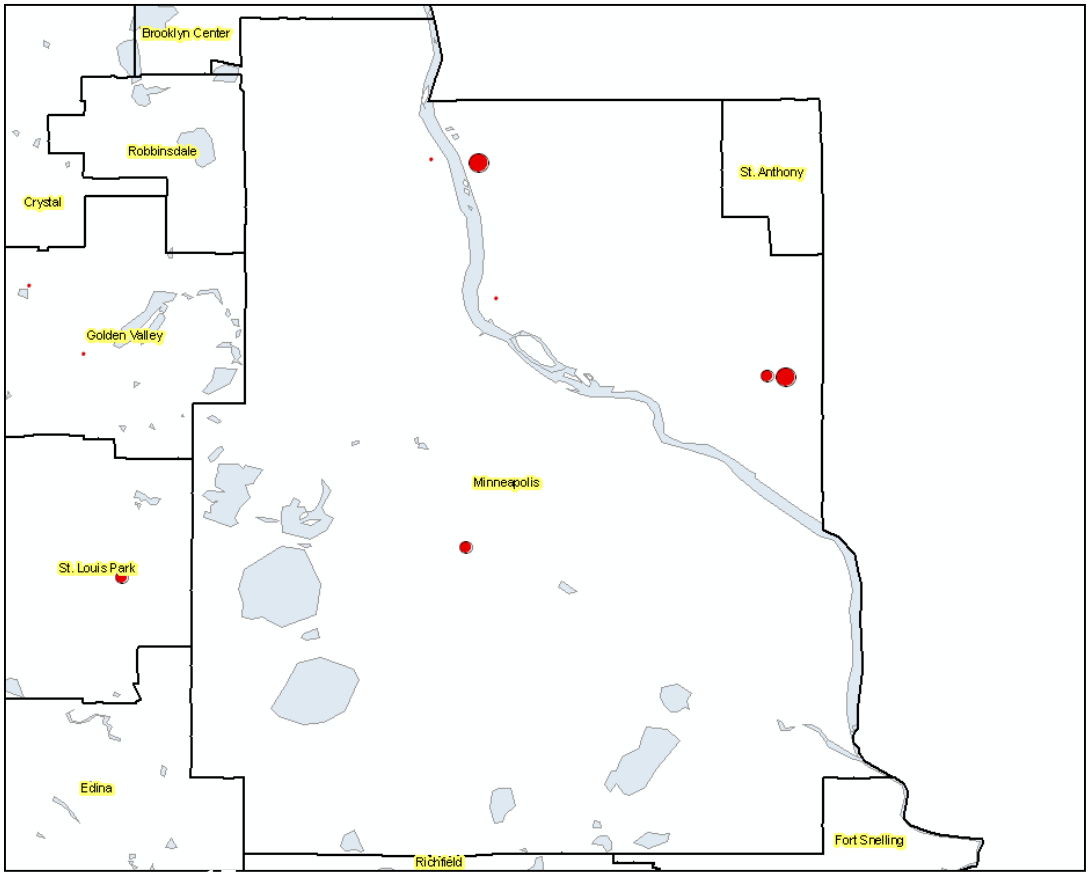
Proximity Analysis

- Proximity is another of the most basic GIS questions, such as:
  - How close is this well to a landfill?
  - Do any roads pass within 1000 meters of a stream?
  - What is the distance between two locations?
  - What is the nearest or farthest feature from something?
  - What is the distance between each feature in a layer and the features in another layer?
  - What is the shortest street network route from some location to another?

Advanced GIS functions

Proximity selections

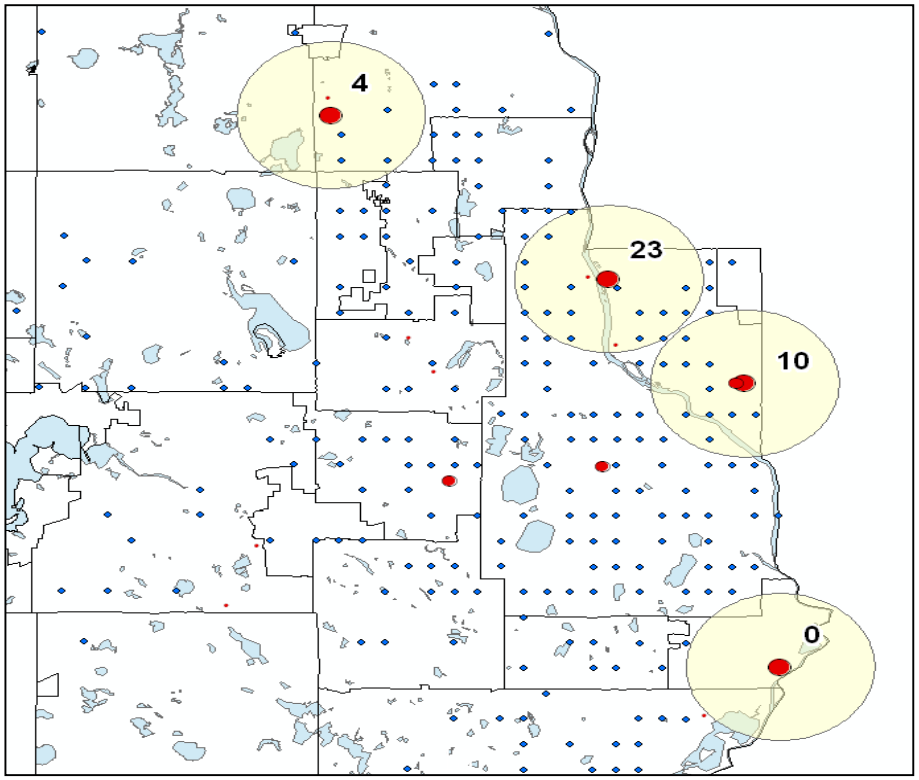




Advanced GIS functions

Buffers

– Select top polluting companies and show the number of schools within 2 miles of these companies.



GIS APPLICATIONS AND EXAMPLES

GIS applications in Planning

- **Planning has different flavours**

  Economic, urban, rural, land use planning

- **Spatial, urban and regional planning**

  "Critical thinking about space and place as a basis for action or intervention"

  "…methods used by the public sector to influence the future distribution of activities in space with the aim of creating more rational territorial organisation of land uses and linkages to balance demands…" (EC 1997)

- **Urban management**

  Day-to-day operation tasks and administration, i.e. construction and street building, parcel and property management, tax collection

GIS applications in Planning

Early ideas: support & improve rational planning

- Driven by research
- Webster (1995), Wegener (1998); Masser, et al (1996)
- Impact of changes in planning, GIS technology

Today: Use of GIS in local & regional government

- For what/why use GIS?
- Is it used for spatial planning, economic planning, urban management?
- How is it used, set up and applications? By whom?

Teaching

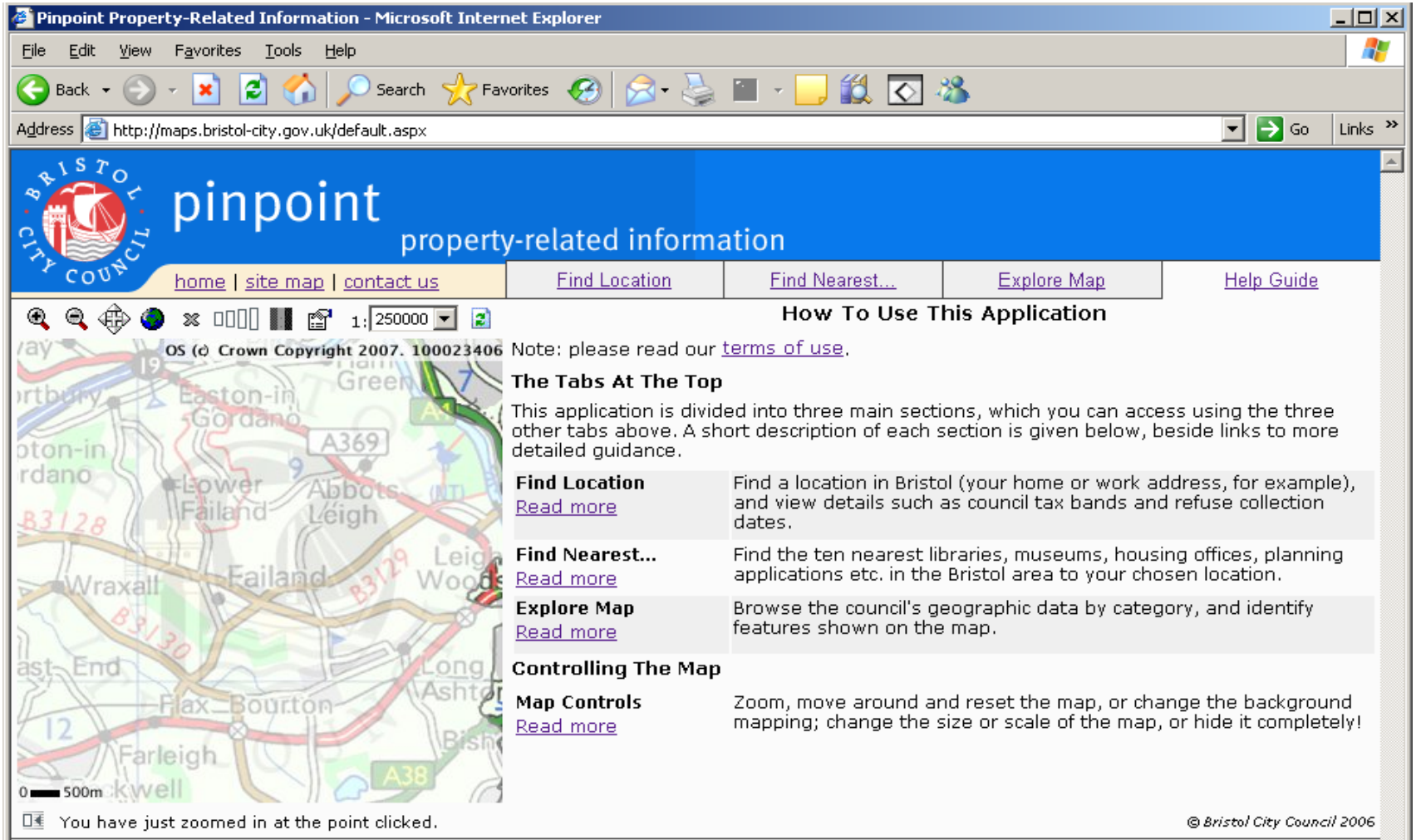
- Building of a GIS
- Mastering Software
- Basic: introductory analysis, mapping
- Advanced: modelling

Implications for GIS education

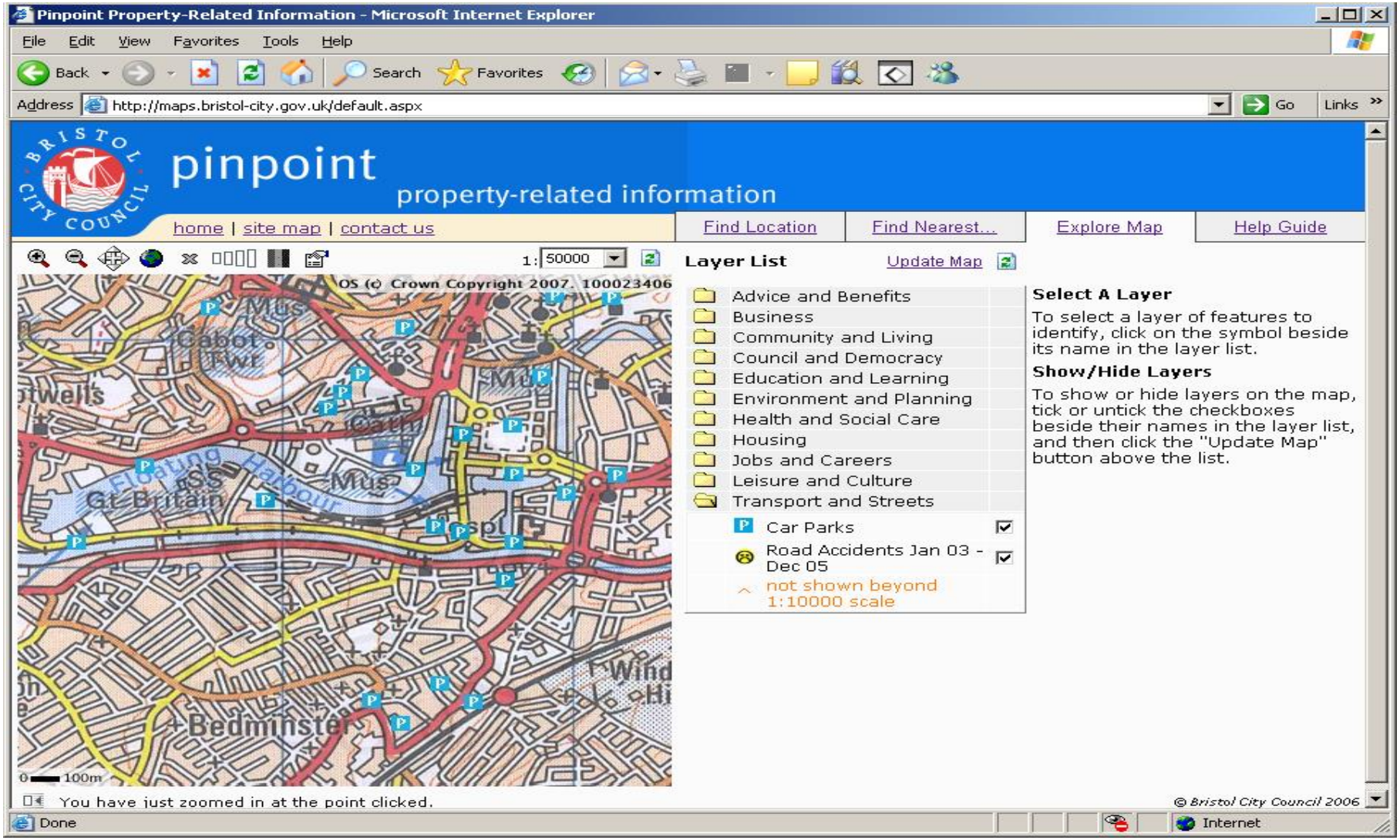
???

Case study 1: Bristol

1. Informing the public ….



With local information ….

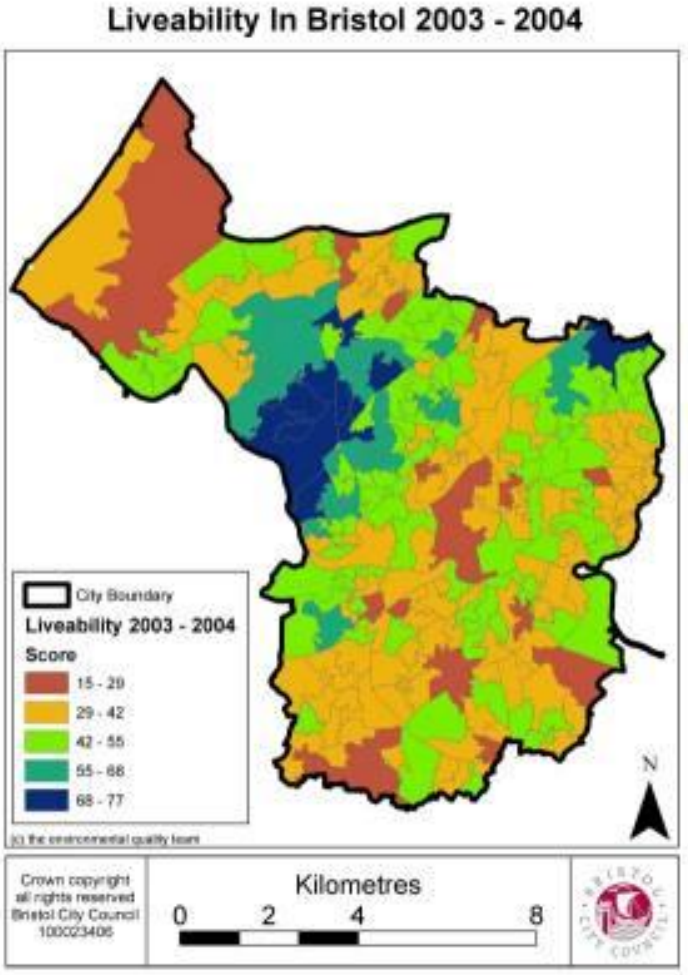


Case Study 1: Bristol

2. Visualisation of data

- Mapping of liveability scores per neighbourhood based on a quality of life survey…..
- To provide clear, timely and location specific evidence for **monitoring** and **review**

3. Emergency planning (GIS is used to access vital information on vulnerable populations and traffic patterns to assist emergency services to operate efficiently)



Case study 2 : Wales

- Welsh Spatial Plan
  - Information and mapping
  - Schematic level illustrations

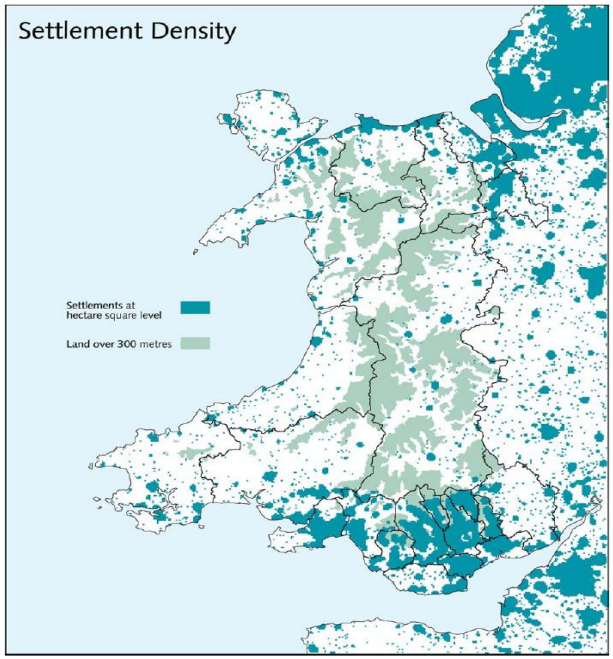


Figure 4 -Settlement Density (presence of residential dwellings by square hectare)

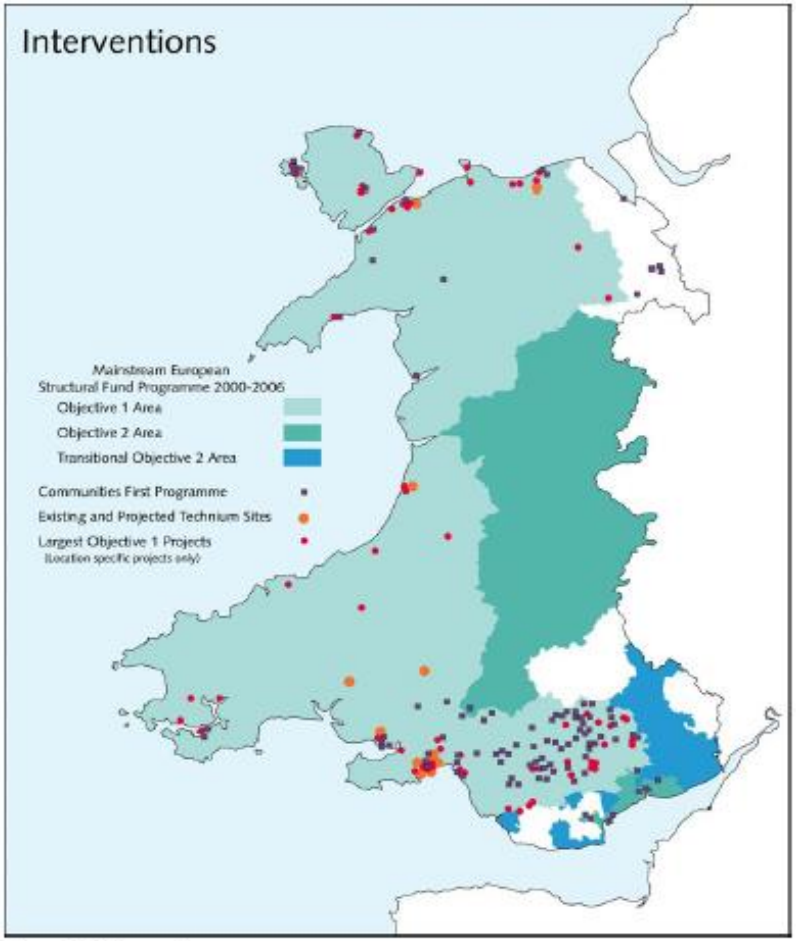
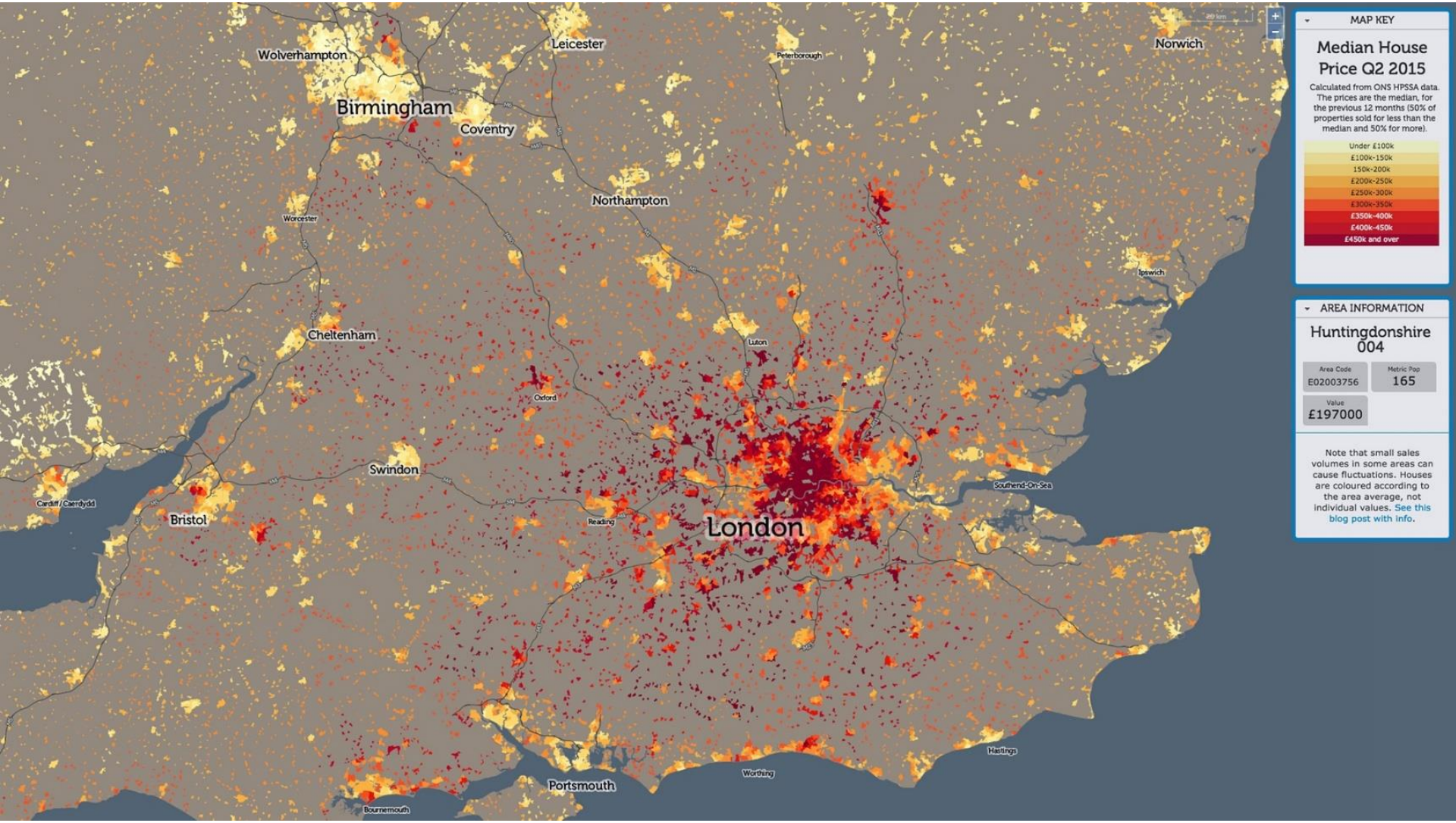
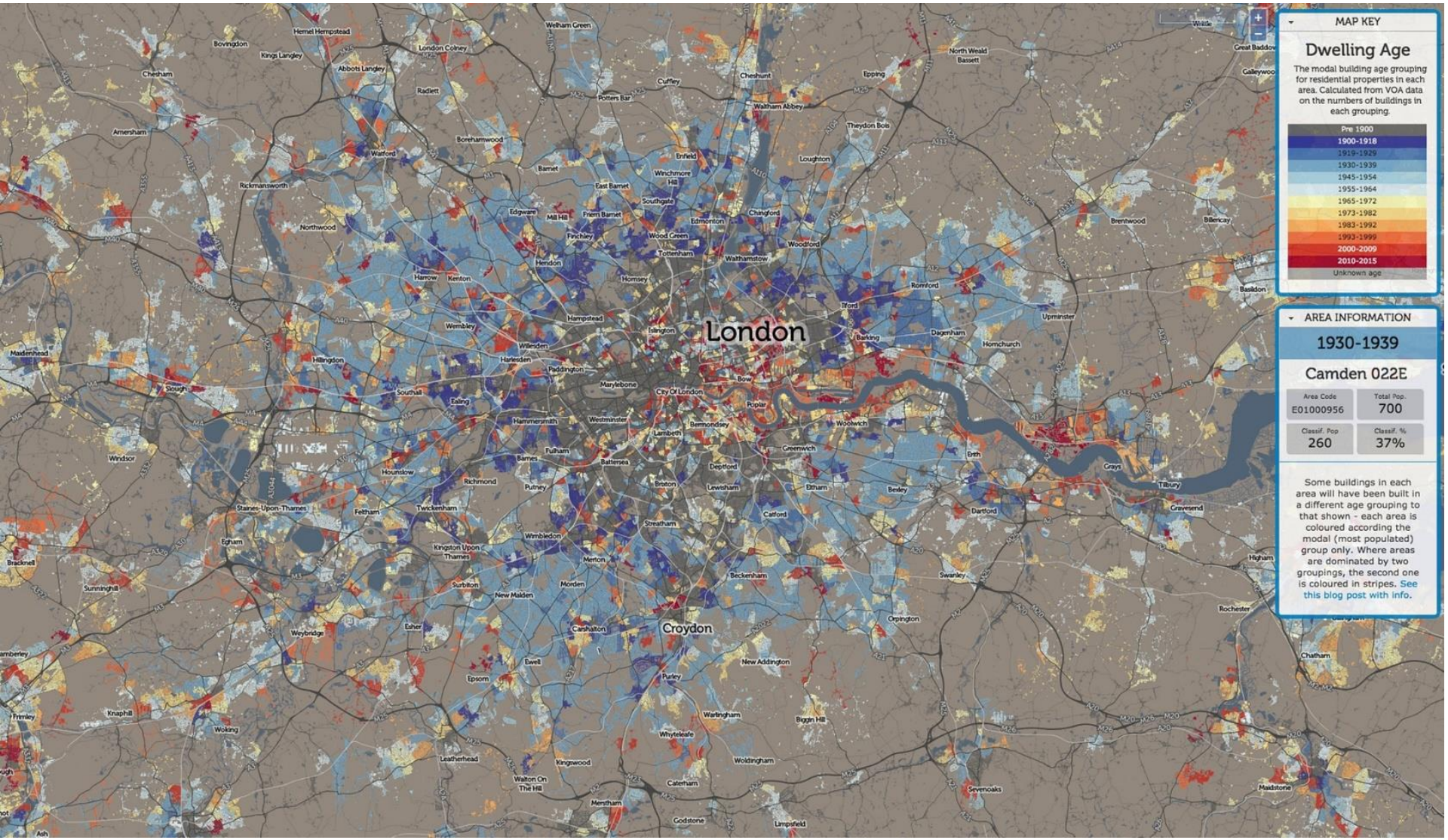


Figure 9 - Interventions

Housing Prices

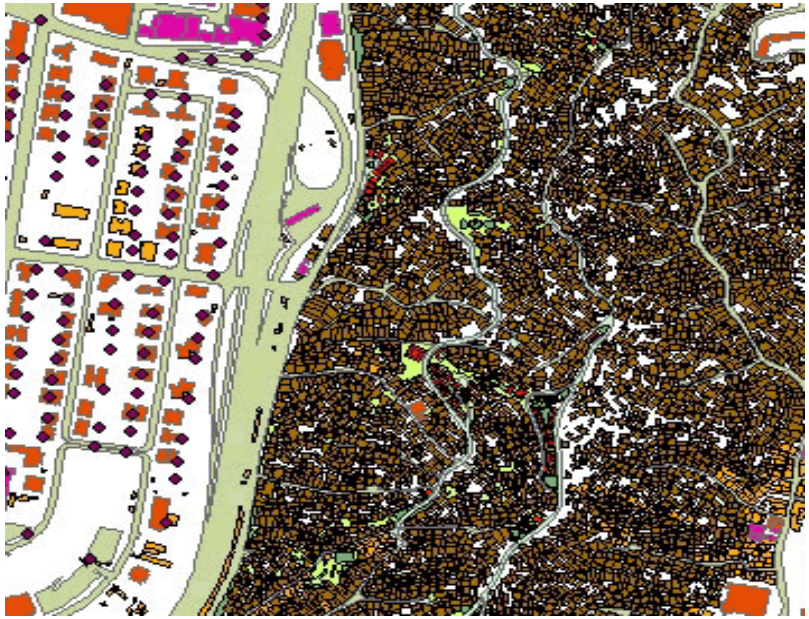


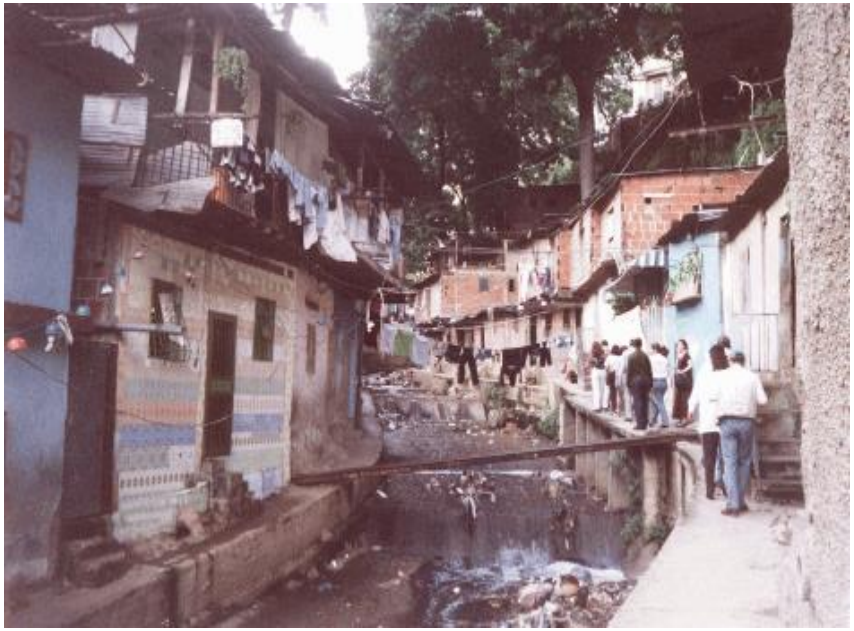
Building Age



GIS for planning underdeveloped areas

Urban poverty measured in terms of quantity and quality of public space.





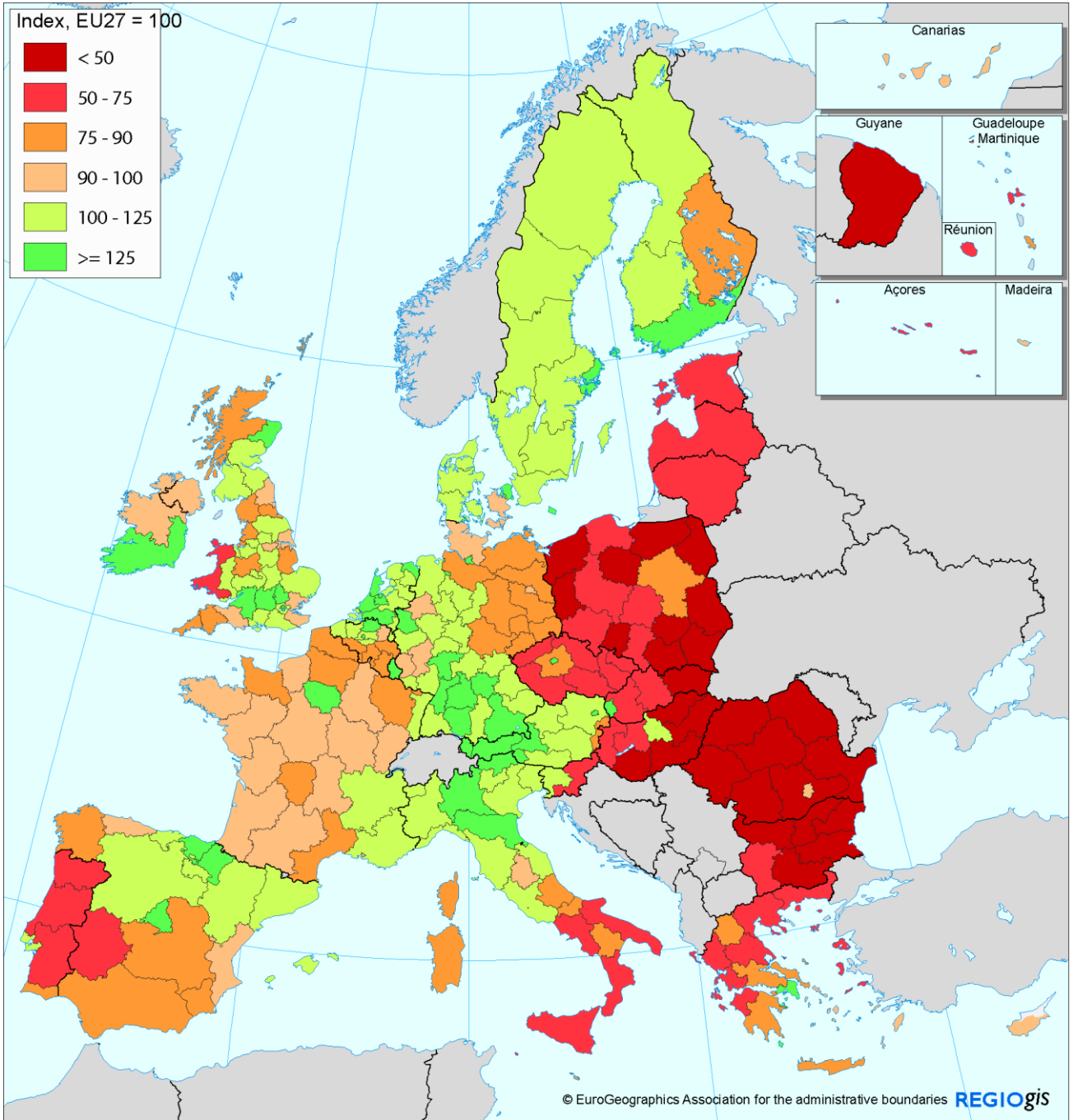
The lack of public open space.

Barrios have a percentage of public space between 5% and 10%.

In the average city total space constitute over 30% of the total space.

The absence of adequate infrastructure, Urban furniture and maintenance which combined produces unhealthy and insecure conditions.

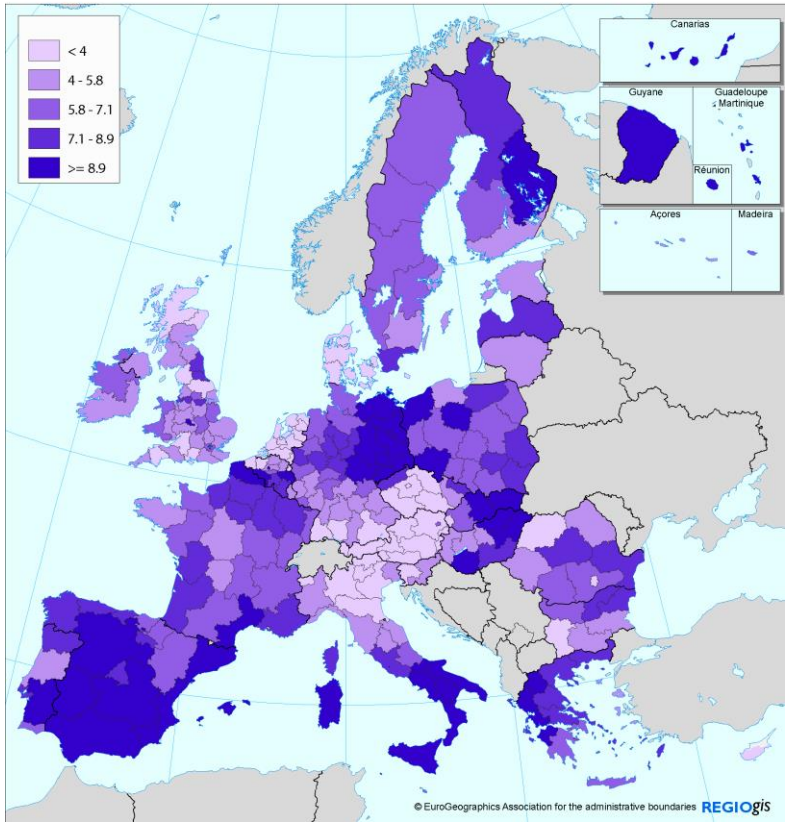
GDP/head (PPS), 2007



Unemployment dropped in most regions until the crisis hit

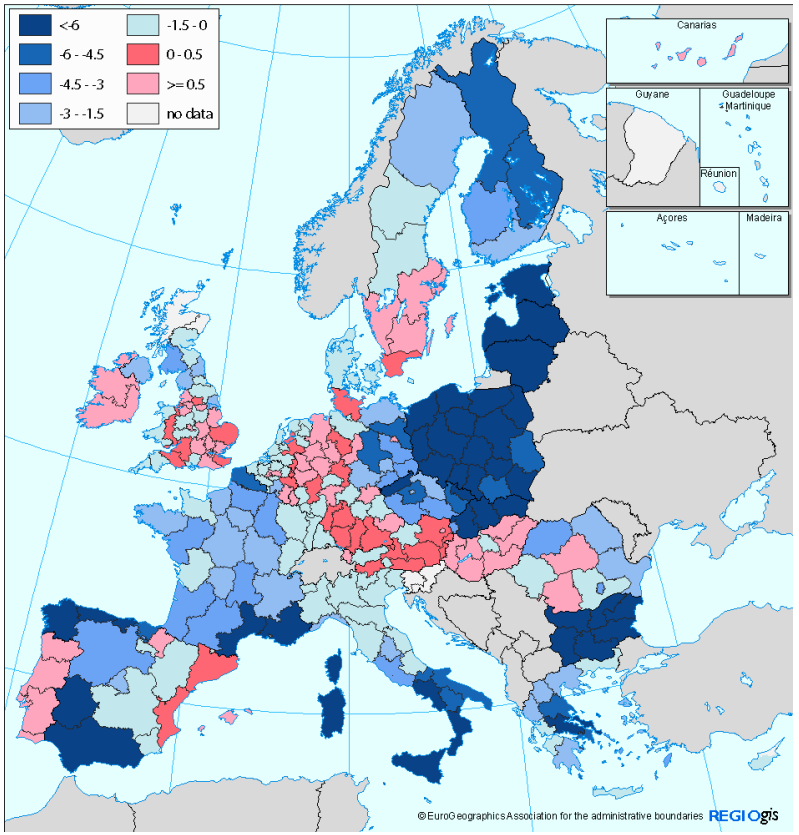
Unemployment rate, 2008

% of active population

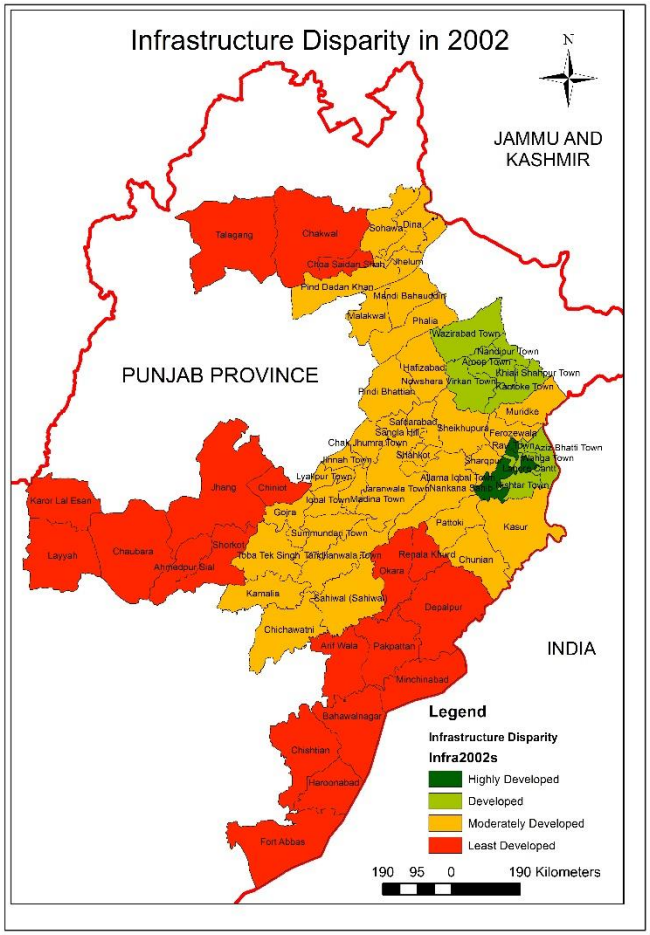


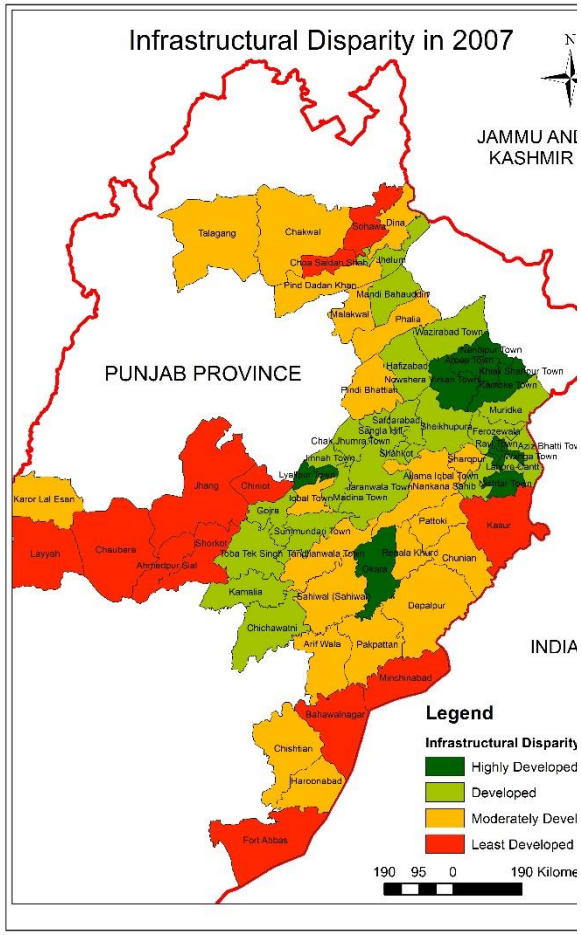
Change in unemployment rate, 2000-2008

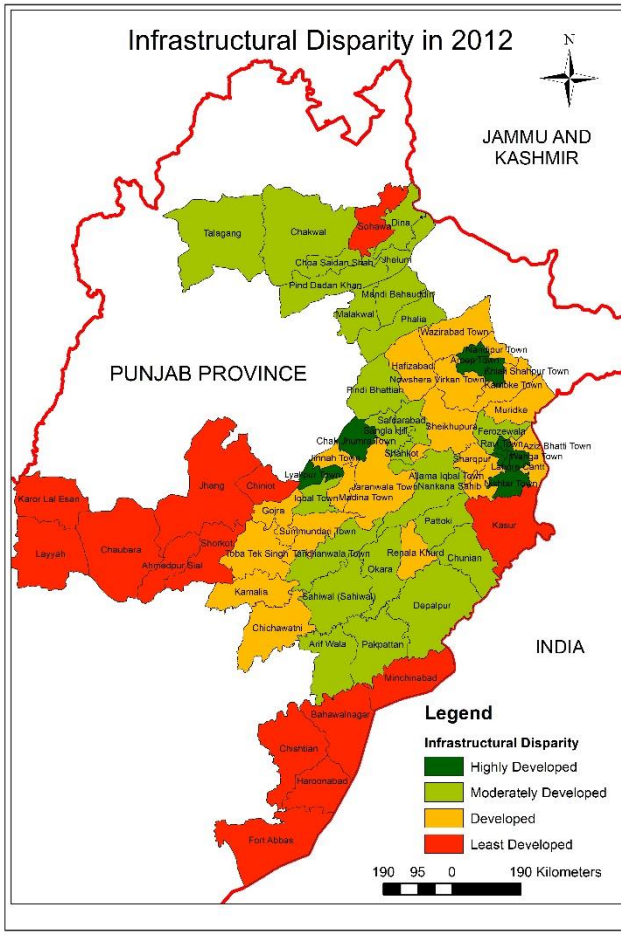
Percentage points change



Infrastructural Development Disparity in Lahore City Region



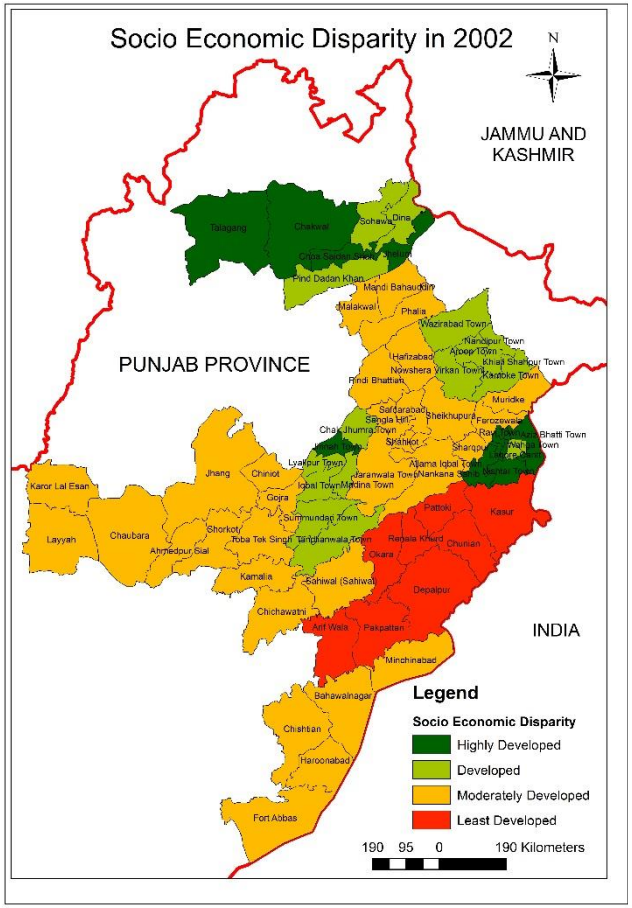


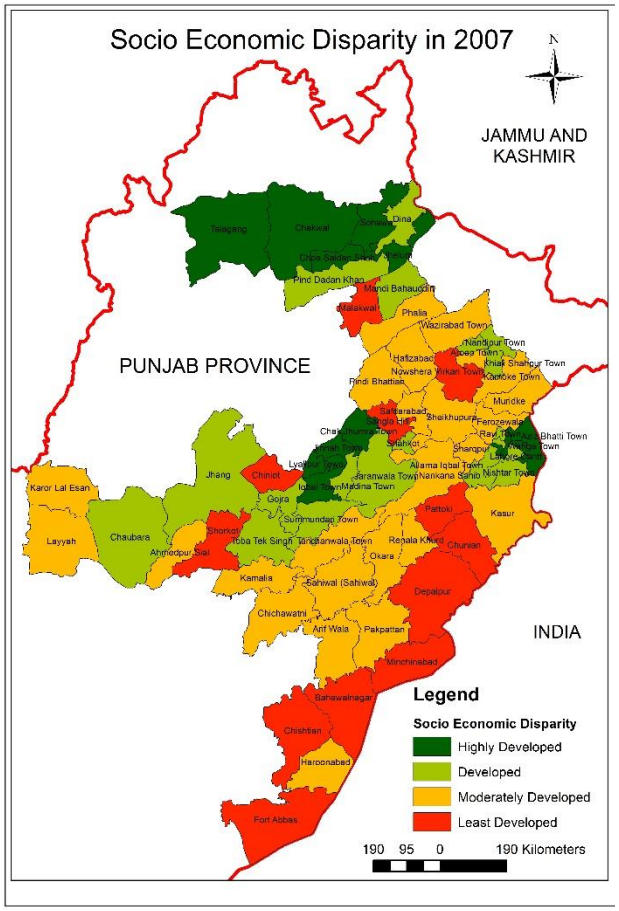


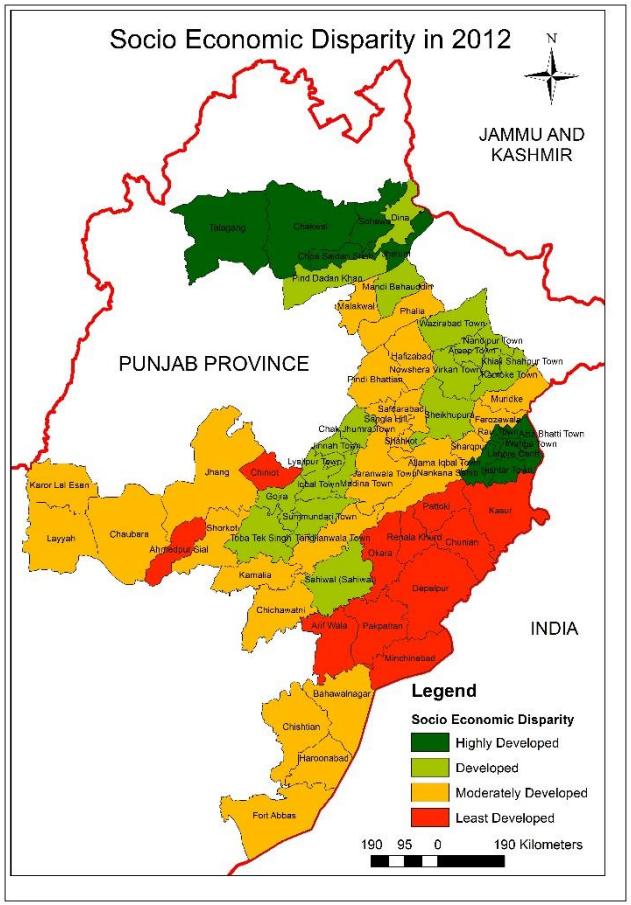
Legend

Highly Developed | Moderately Developed | Developed | Least Developed

Socioeconomic Development Disparity in Lahore City Region







Legend

Highly Developed

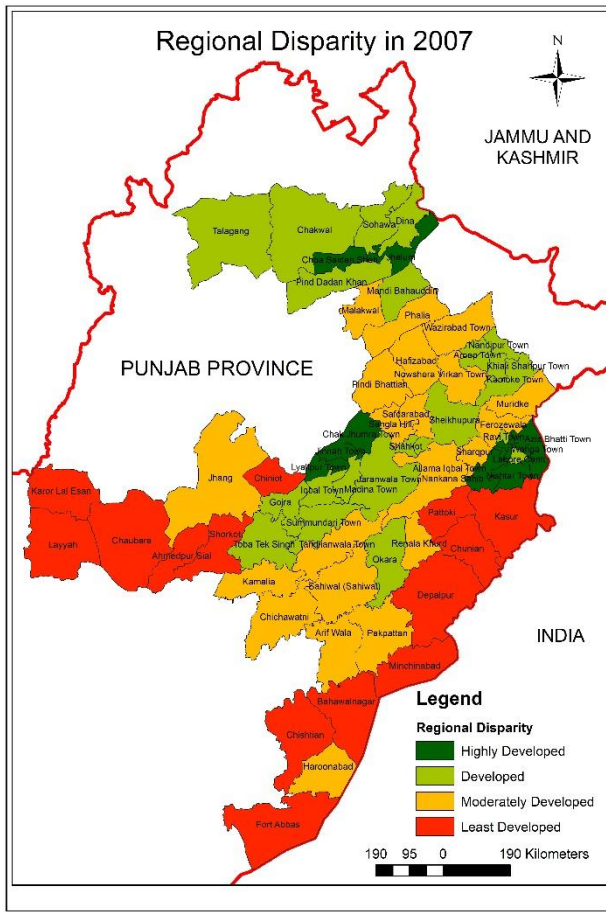
Moderately Developed

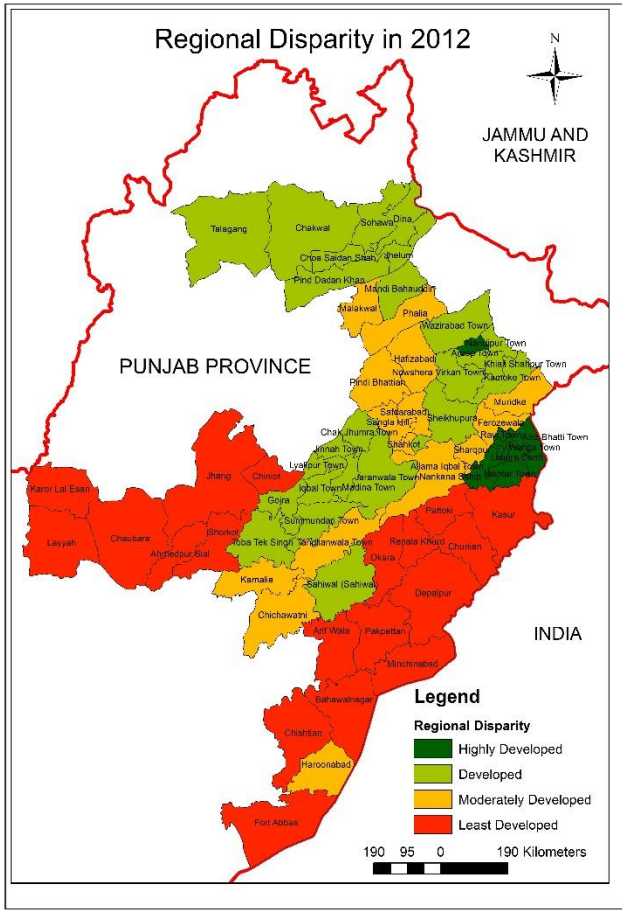
Developed

Least Developed

Regional Development Disparity in Lahore City Region







Legend

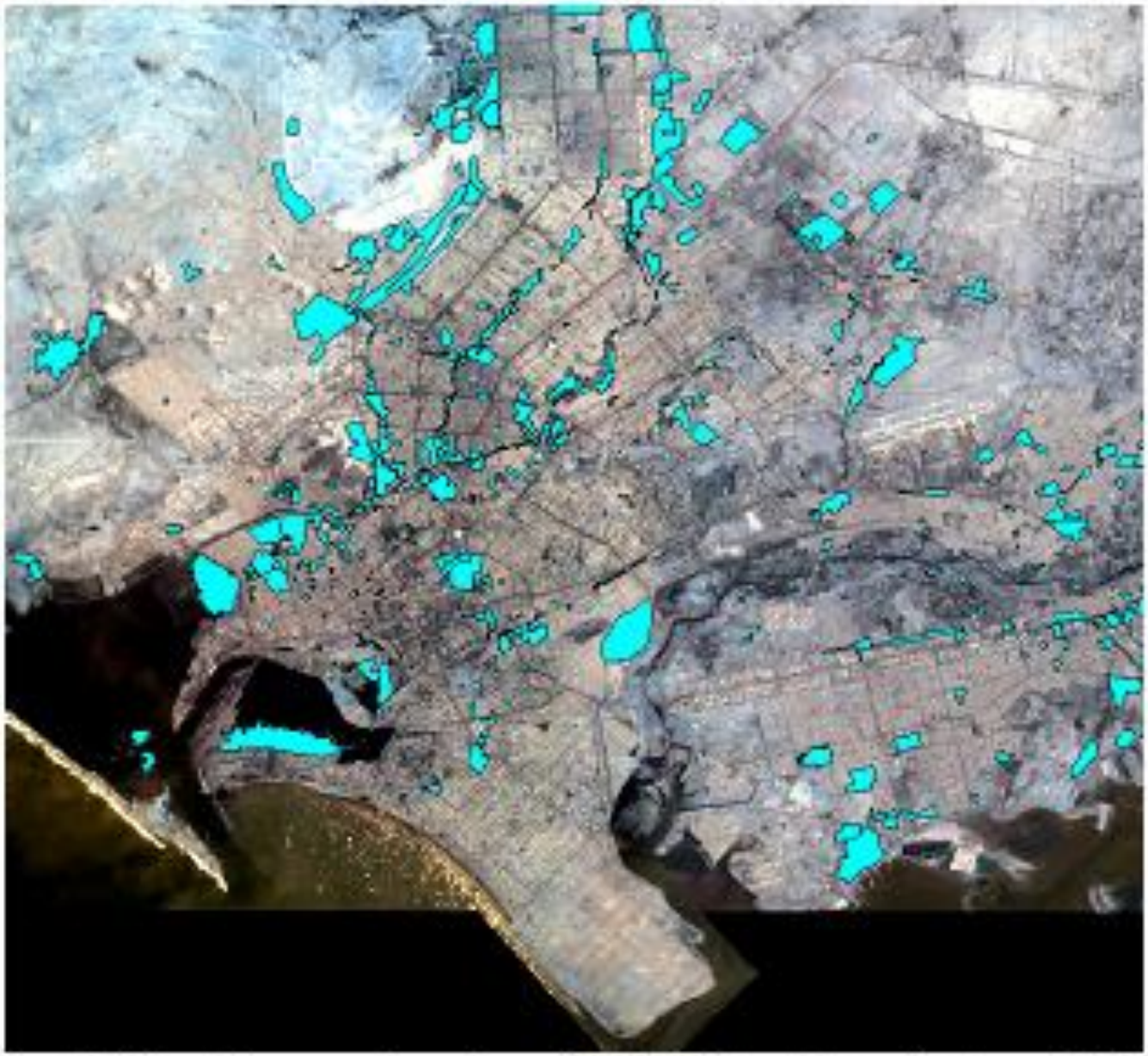
Highly Developed | Moderately Developed | Developed | Least Developed

Quality of Life

- **Computed using:**
  - **Physical health**
  - **Psychological**
  - **Social relationships**
  - **Environment**
  - **Economic condition and development**
  - **Access to facilities and services**



Bhatti et al. (2016). Social Indicators Research



Katchi Abadis marked on Satellite data (Year 2005)

Aspects and Impacts of Urban Planning

- Physical and Economic aspects (requires environmental , land ownership; physical infrastructure; and topographic data

- **Social impact** (sometimes is overlooked due to lack of socio-economic data).
  - The trend of urban expansion to suburb area (far from the city center) can cause social problem such as: difficulties in accessing the working area and lack of public transportation.

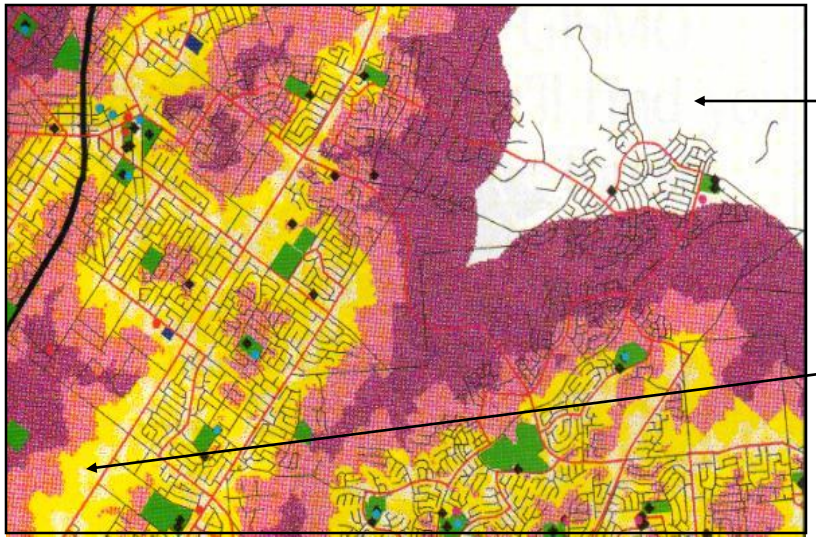
Socio-Economic Data

- **Services and Facilities data** (education, health, child care, emergency services, mosques/churches, recreation, public transport);
- **Land Use data** (current land use, open space, industrial locations, retail locations);
- **Population data** (demographic characteristics, population projections);
- **Land and Housing data** (# of dwellings, age and type of dwellings, available allotments, broad area land, forecast allotment demand).

Spatial Unit Data

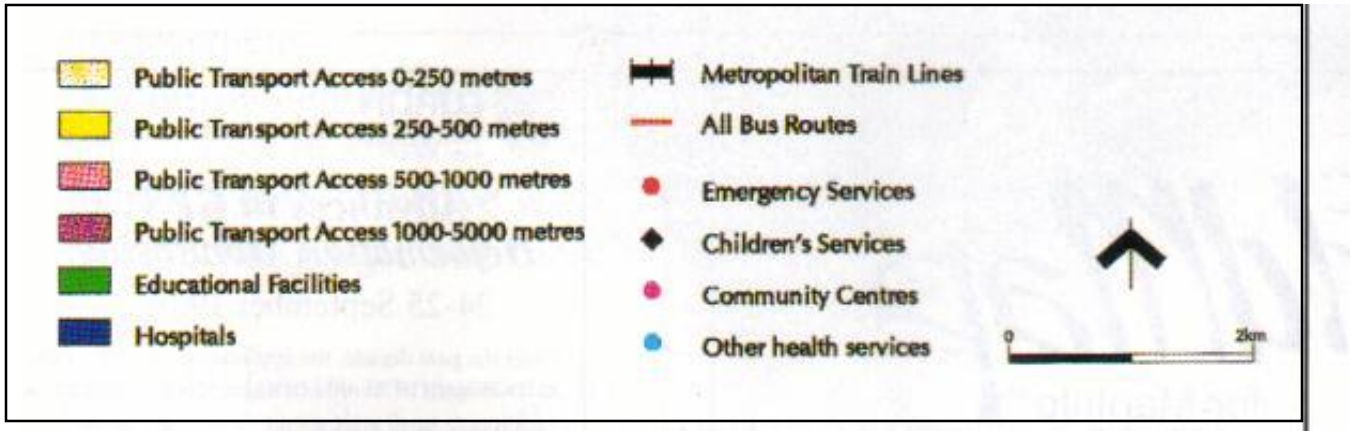
- Provide data at the level of individual land parcel;
- So that it can present information from the detailed local community level through the analysis of city-wide planning issues.

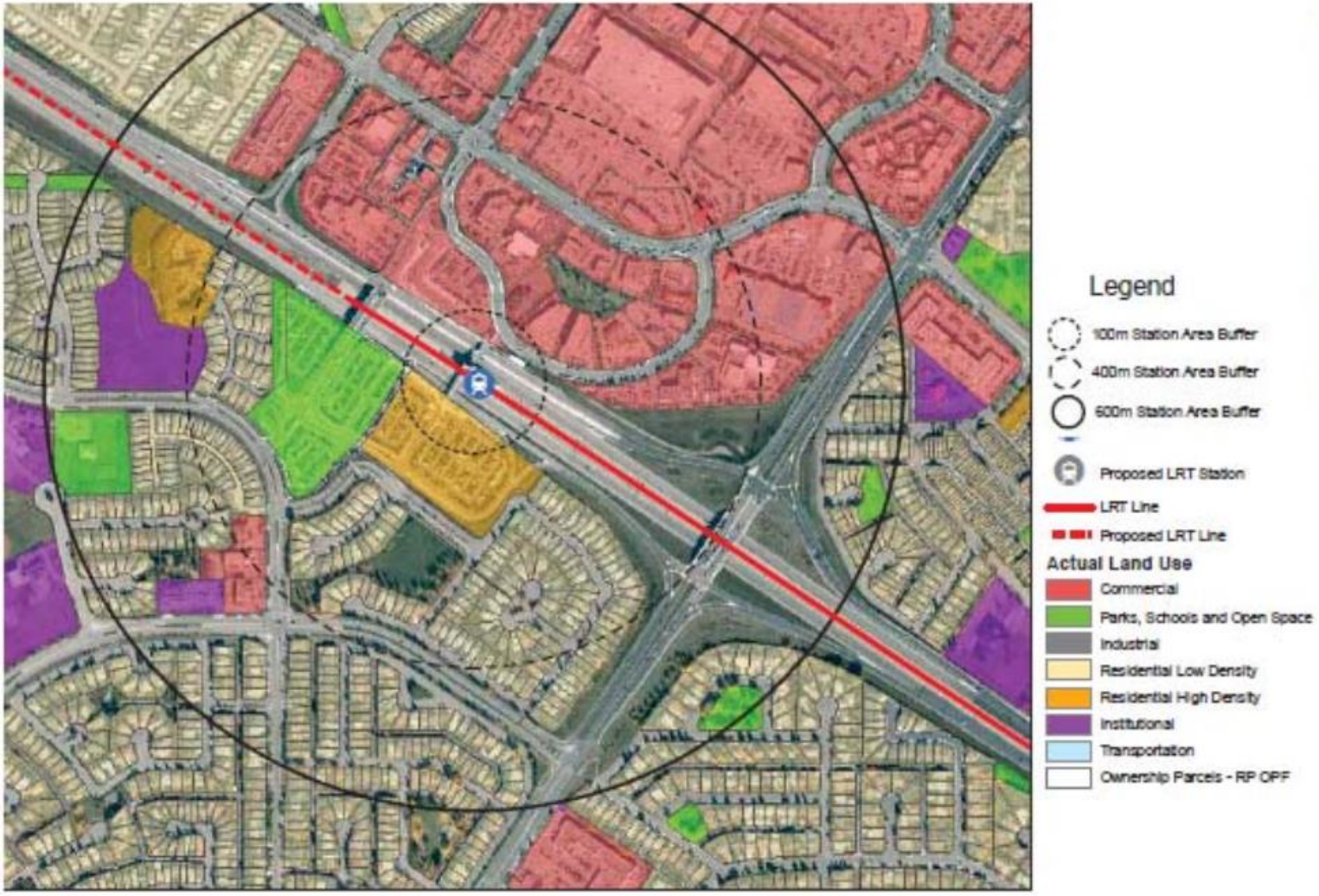
ASimple Map of the Facility Locations



Lack of Public Transport Access

Public Transport Access within 0-250 meters





Acutal Land Use from Land Use - Property Use

Crowfoot, Calgary, Canada

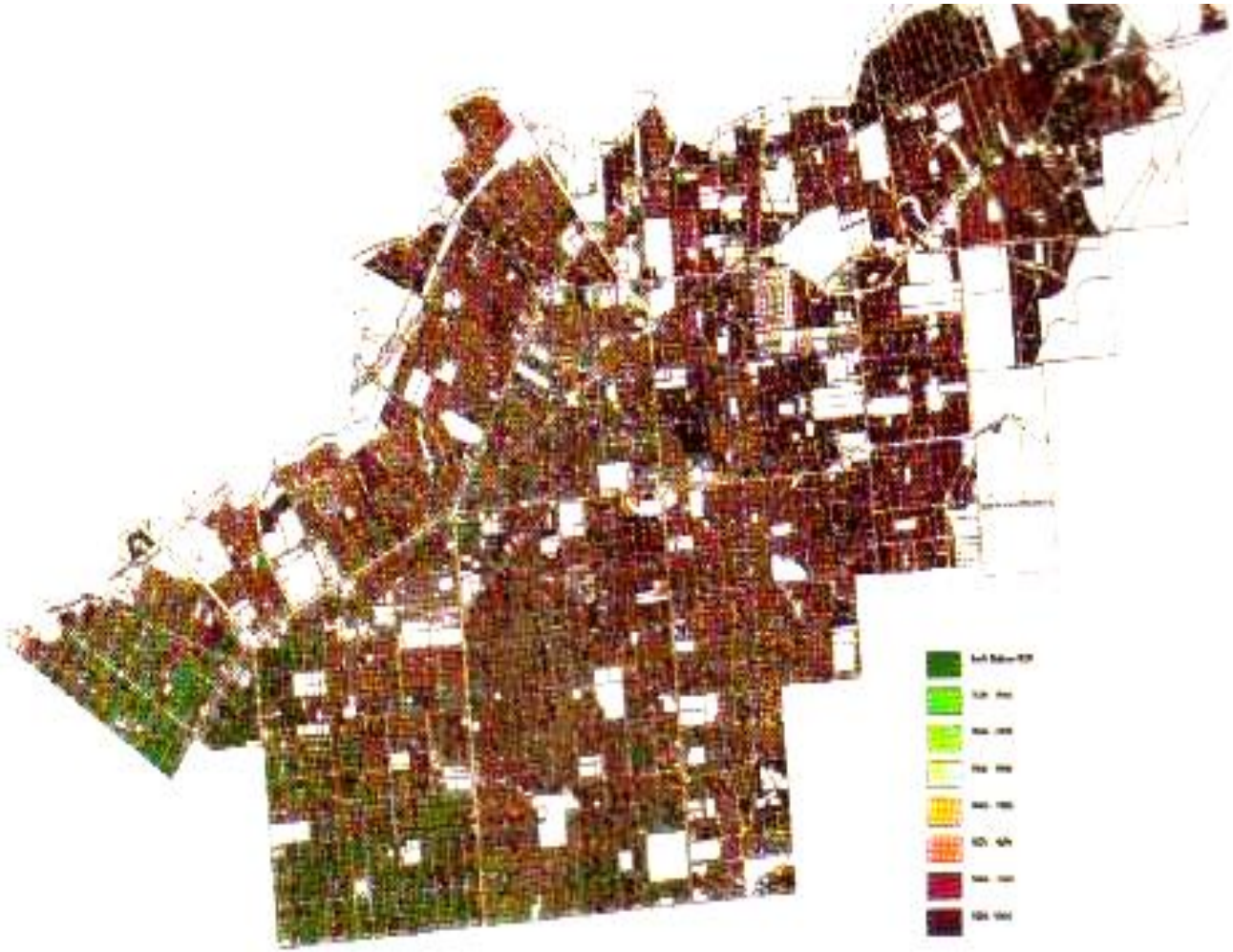
Problems versus Solutions in Urban Planning

- *Major Problems*
  - The development of new residential areas requires the building of new facilities (schools, hospitals) and basic physical infrastructures (roads, water, electricity);
- *Key Issues*
  - What will be the future demand for services (population and demographic forecast);
  - Where should these services be located;
  - When should these services be built;
- ***Solutions Require the Following Information***
  - The current population, the locations of current services and their capacity, the estimation of the future population number and distribution.

Applications of Urban Planning Database

- Analysis of development trends;
- Analysis and monitoring of land and housing markets;
- Development of regional strategic plans;
- Development of community plans;
- Analysis of school bus transport systems;
- Modeling of accessibility to public transport.

Residential development pattern

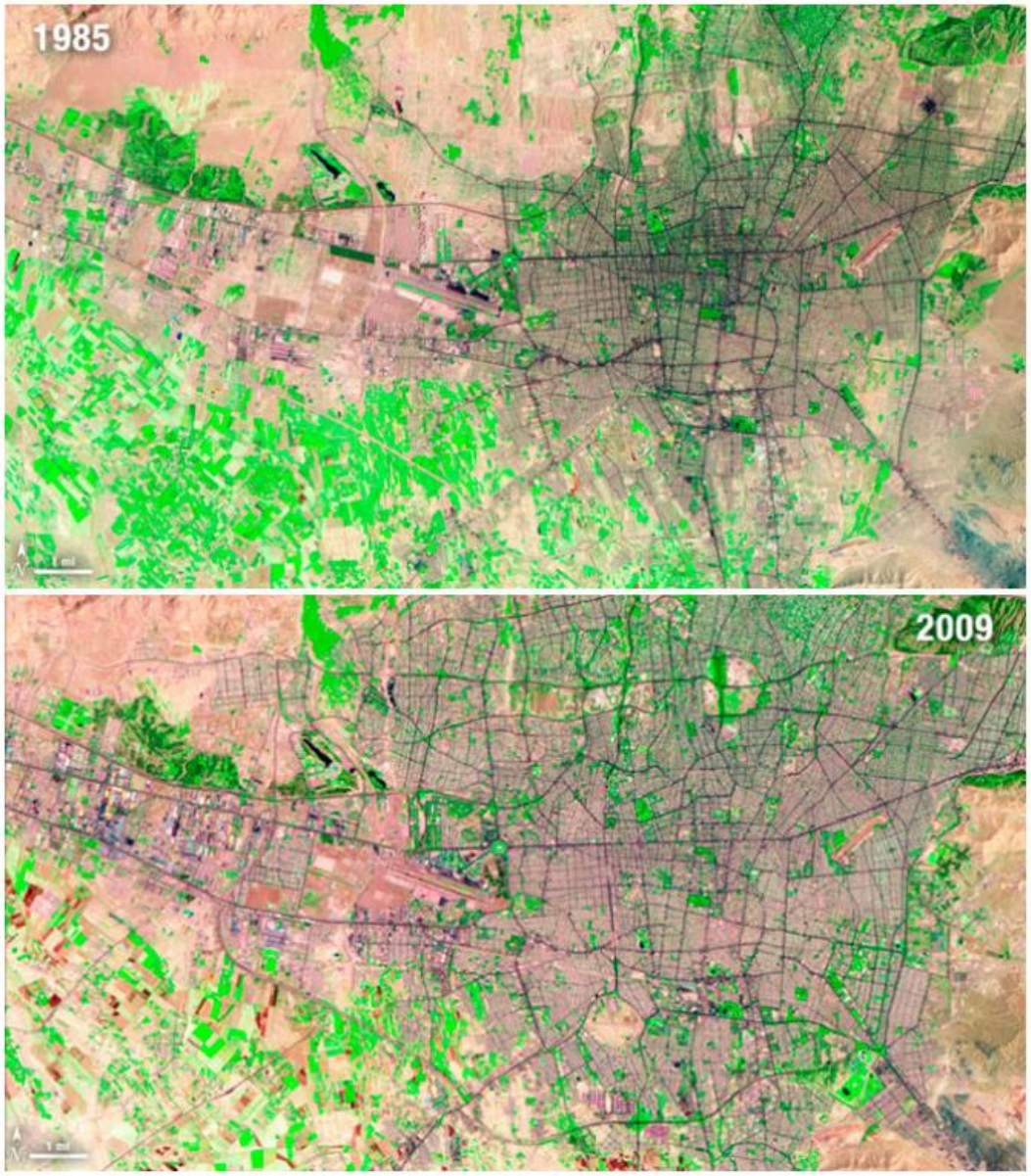


Green to dark blue was built a long time ago;

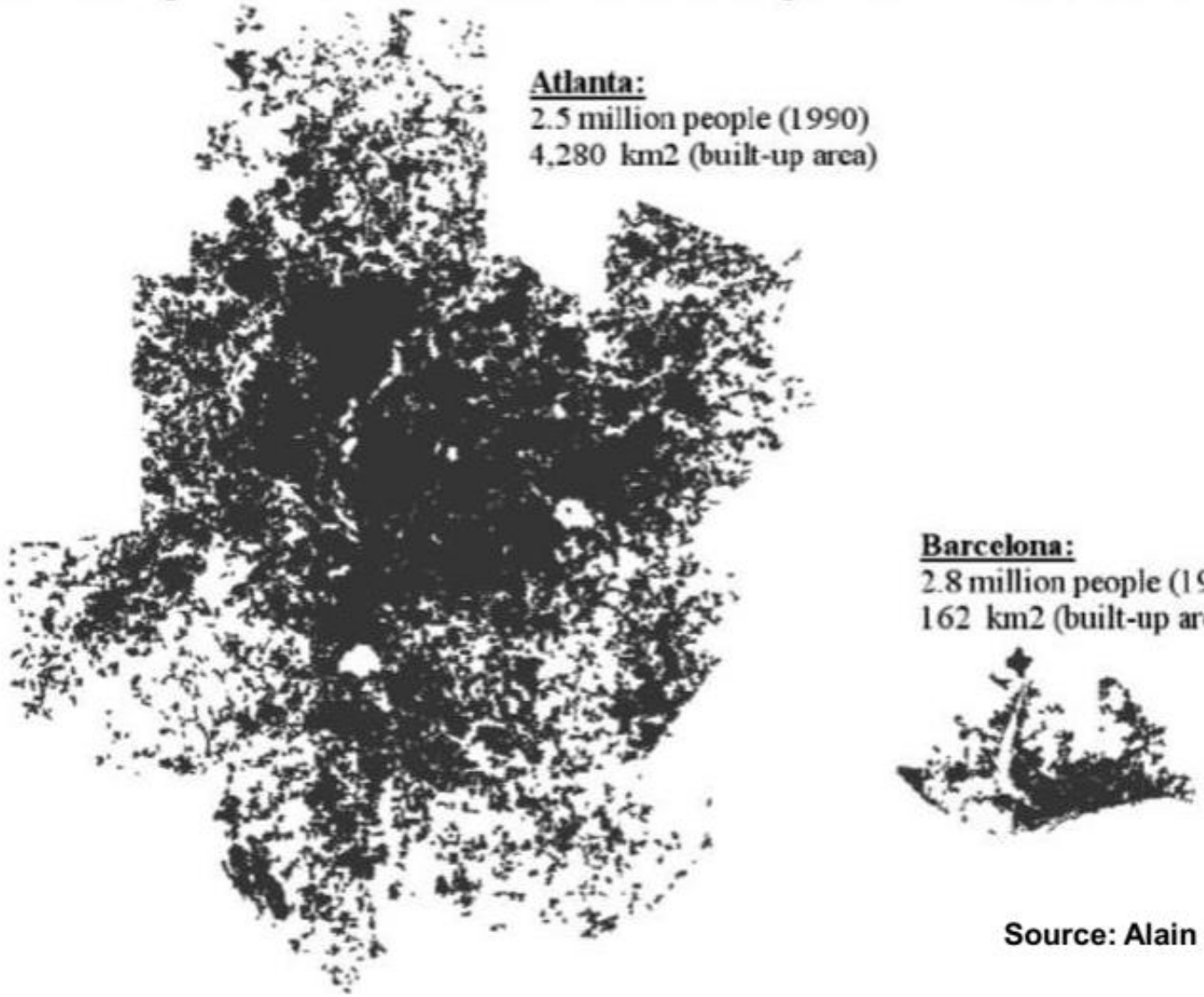
Red to dark purple was built more recently.

Urban Growth

Tehran



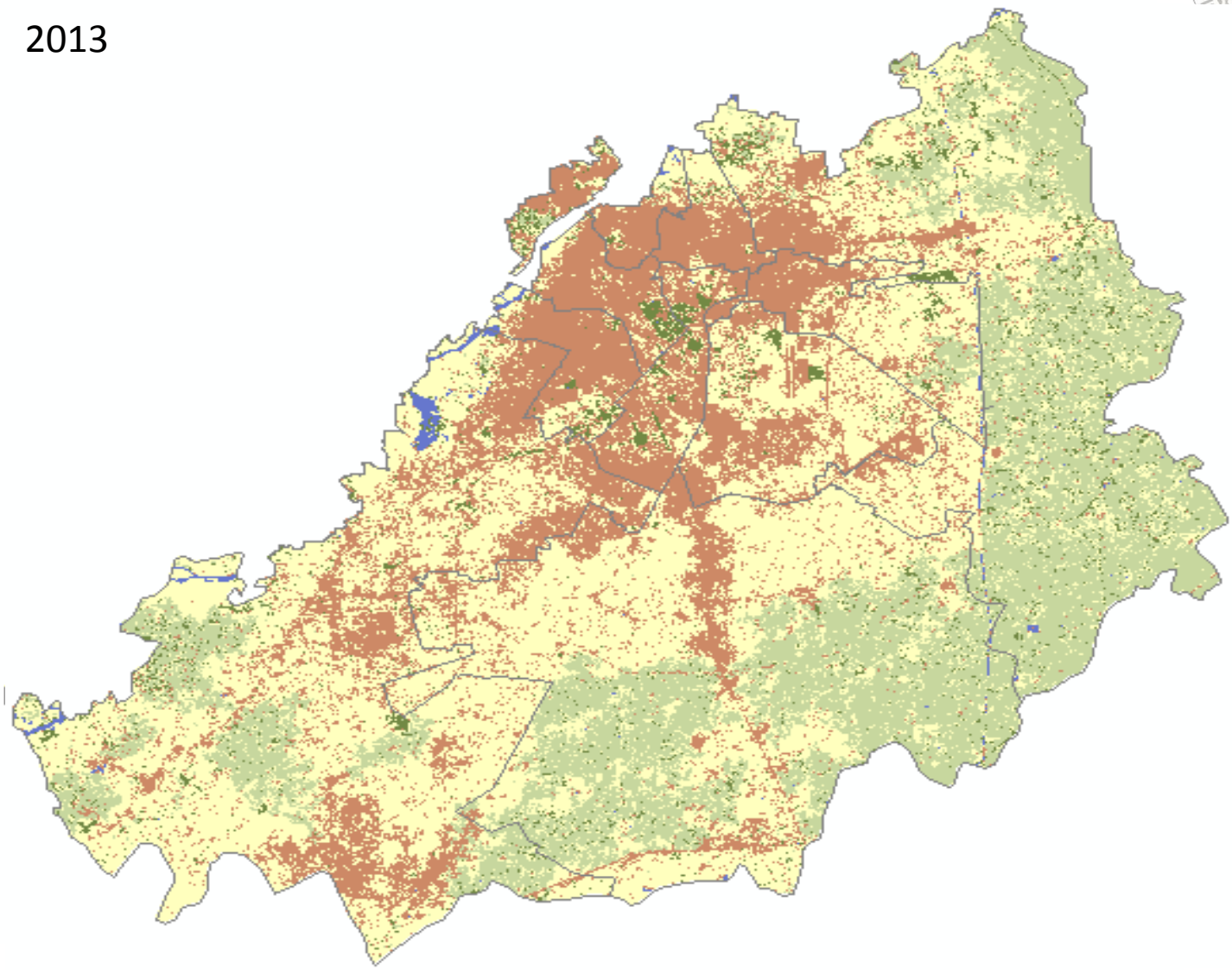
The Built-up Area of Atlanta and Barcelona Represented at the Same Scale



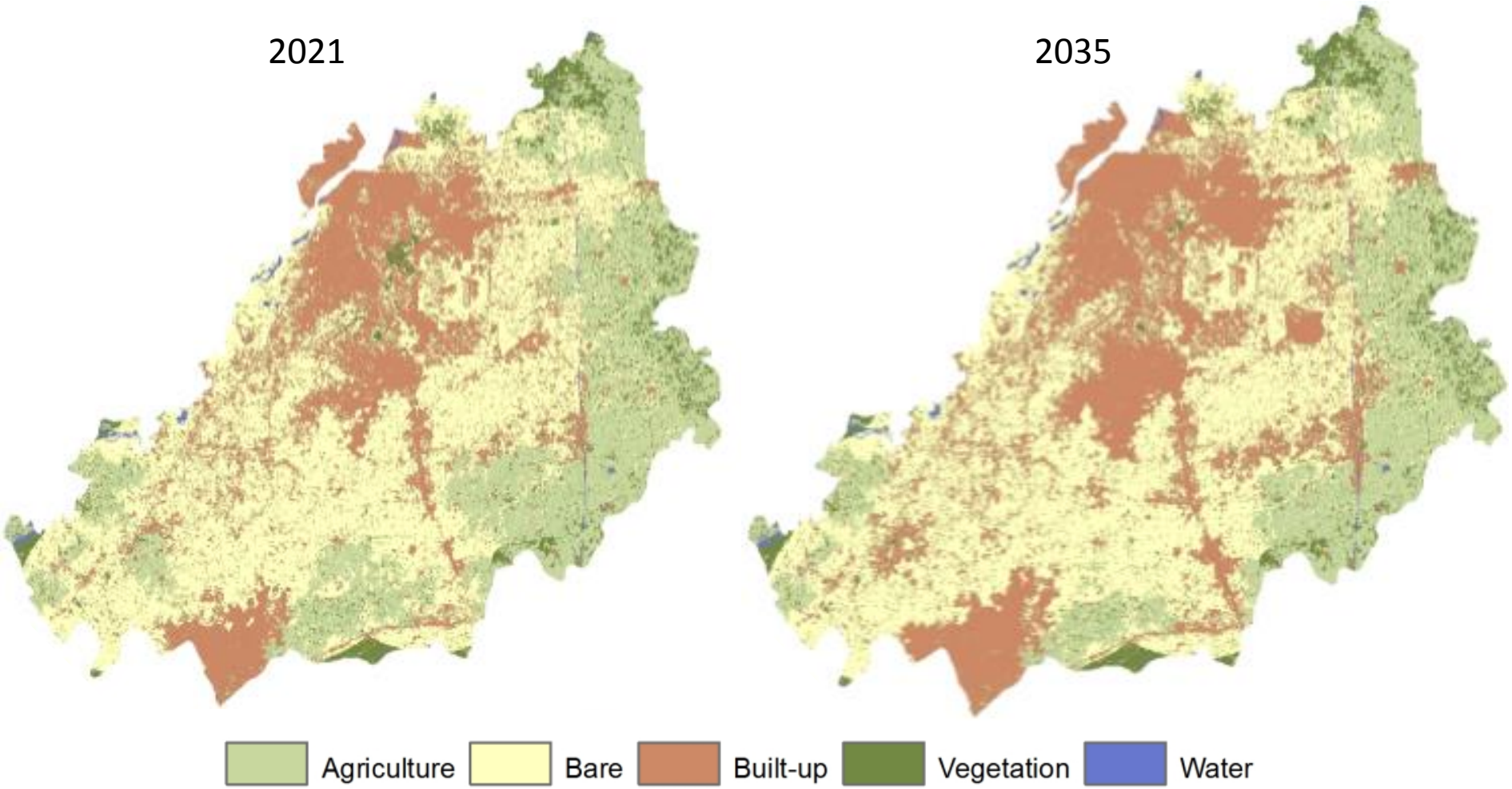
Source: Alain Bertaud, 2002

Lahore

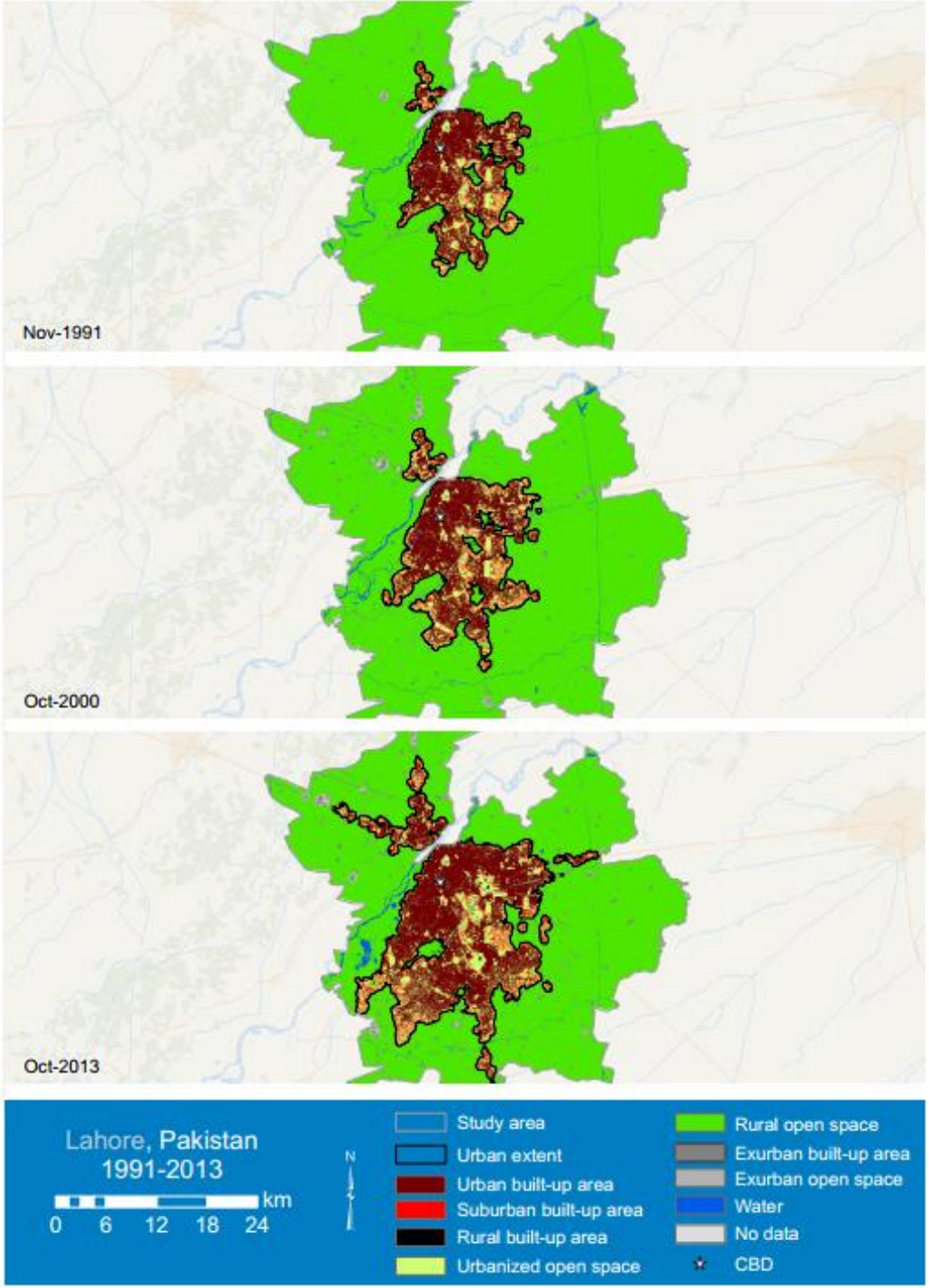
2013

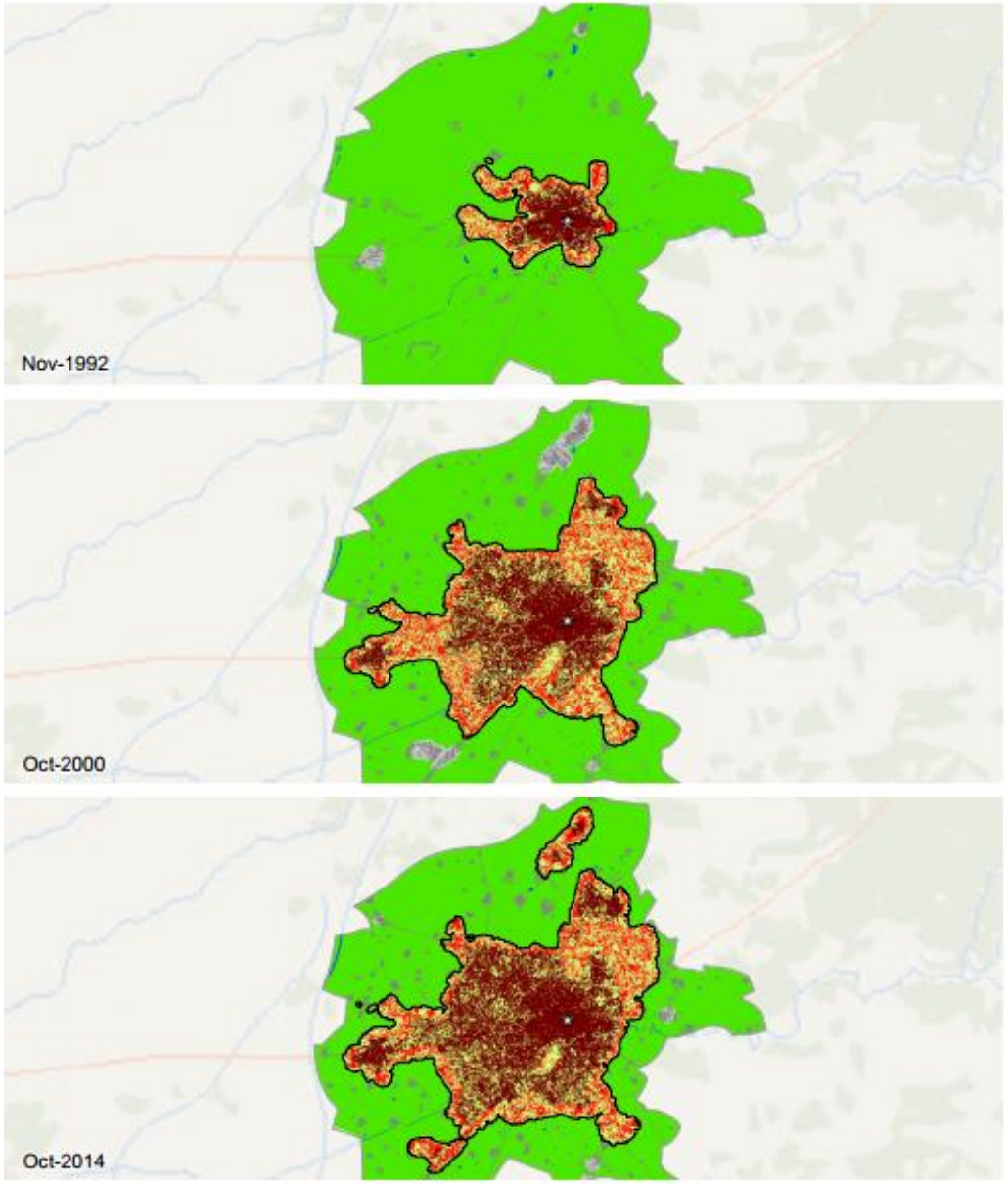


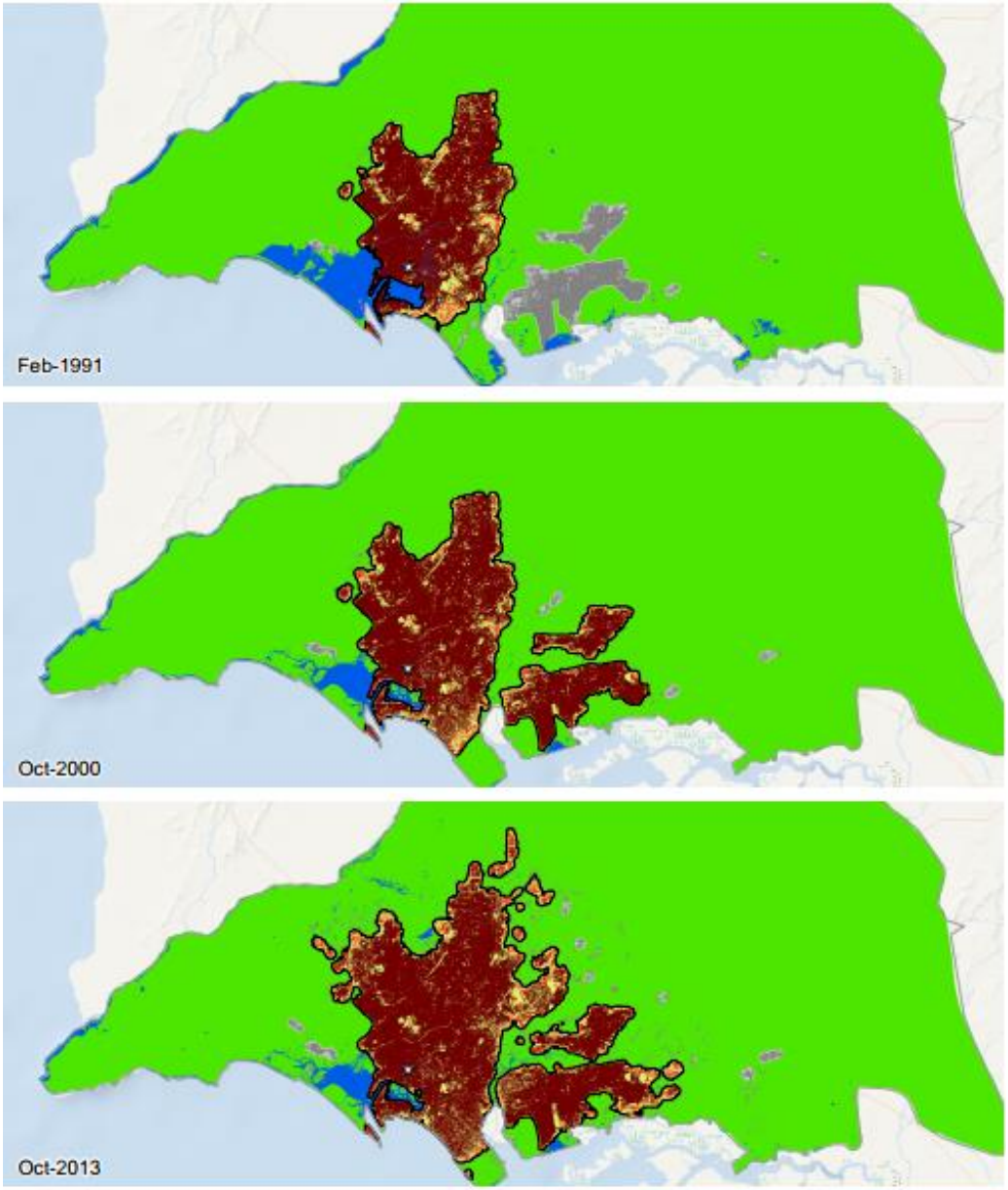
Prediction



Source: Bhatti …. Rana et al., Habitat International 2015


Nov-1991
Oct-2000
Oct-2013
Lahore, Pakistan
1991-2013
km
0 6 12 18 24
N
Study area
Urban extent
Urban built-up area
Suburban built-up area
Rural built-up area
Urbanized open space
Rural open space
Exurban built-up area
Exurban open space
Water
No data
CBD


Nov-1992
Oct-2000
Oct-2014
Sialkot, Pakistan
1992-2014
km
0 2 4 6 8
N
Study area
Urban extent
Urban built-up area
Suburban built-up area
Rural built-up area
Urbanized open space
Rural open space
Exurban built-up area
Exurban open space
Water
No data
CBD


Feb-1991
Oct-2000
Oct-2013
Karachi, Pakistan
1991-2013
km
0 5 10 15 20
N
Study area
Urban extent
Urban built-up area
Suburban built-up area
Rural built-up area
Urbanized open space
Rural open space
Exurban built-up area
Exurban open space
Water
No data
CBD

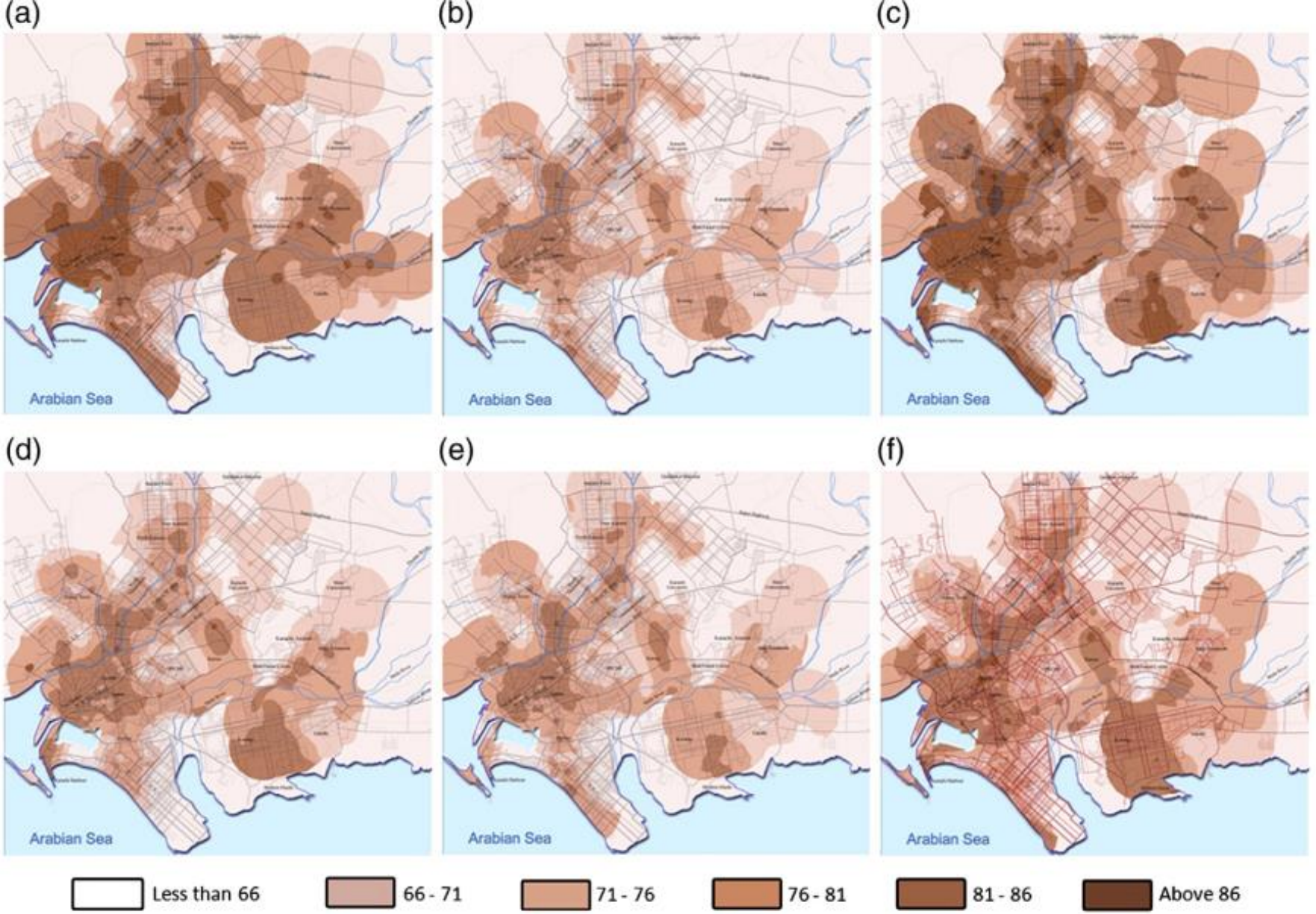
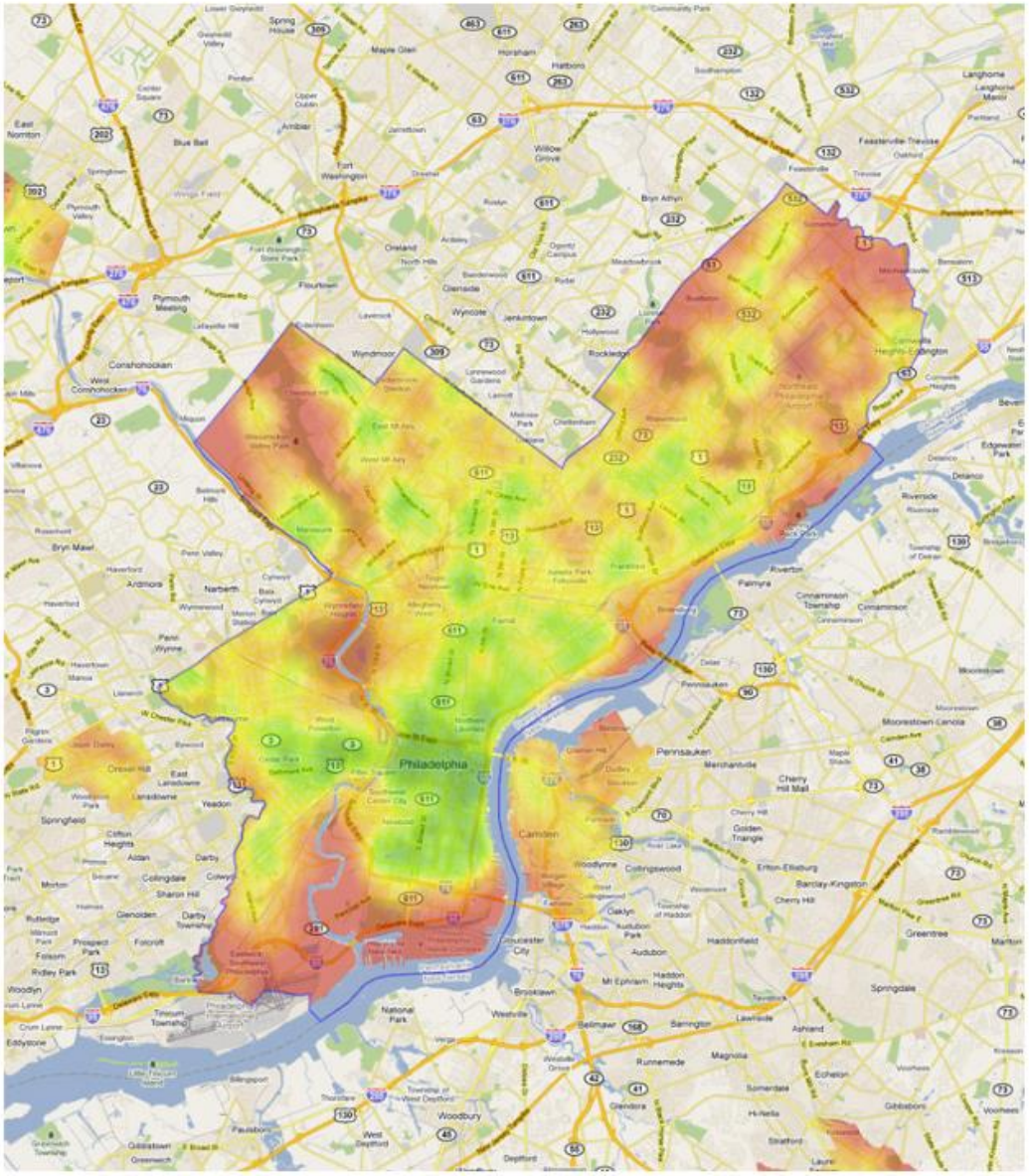


Fig. 5. Mode road traffic noise levels for six given time periods (unit: dB): (a) weekday morning, (b) weekday afternoon, (c) weekday evening, (d) weekend morning, (e) weekend afternoon and (f) weekend evening. Morning time was 06:30 to 10:30, afternoon 12:00 to 15:00, and evening 16:30 to 24:00.

Source: Mehdi et al. (2011) Spatio-temporal patterns of road traffic noise pollution in Karachi, Pakistan. Environment International

This heat map is a visualization of the relative walkability scores of the differen neighborhoods in Philadelphia, where green equates to very walkable and red means less walkable. Click here to explore Philly's walkability on an interactive map.


Philadelphia
Conshohocken
Pennsauken
Camden
Cherry Hill Mall
Moorestown-Lenola
Haddonfield
Greentree
Springdale
Woodbury
Deptford
Delanco
Riverside
Palmyra
Cinnaminson
Bryn Mawr
Ardmore
Wyncote
Jenkintown
Rockledge
Willow Grove
Fort Washington
Plymouth Meeting
Flourtown
Wyndmoor
Oreland
Glenside
Bryn Athyn
Langhorne
Feasterville-Trevose
Bensalem
Edgewater Park
Merchantville
Barclay-Kingston
Runnemede
Magnolia
Somerdale
Haddon Heights
Mt Ephraim
Audubon
Collingswood
Woodlynne
Oaklyn
Brooklawn
Westville
Gloucester City
Pennsauken
Springfield
Yeadon
Lansdowne
Darby
Sharon Hill
Colwyn
Folcroft
Ridley Park
Prospect Park
Glenolden
Norwood
Morton
Rutledge
Media
Swarthmore
Wallingford
Newtown Square
Havertown
Haverford
Wynnewood
Penn Wynne
Narberth
Bala Cynwyd
Merion Station
Villanova
Radnor
Paoli
Malvern
Phoenixville
Valley Forge
Norristown
West Norriton
East Norriton
Plymouth Valley
Blue Bell
Ambler
Jamison
Upper Dublin
Lower Gwynedd
Spring House
Maple Glen
Horsham
Hatboro
Southampton
Langhorne Manor
Oxford
Trevose
Fairless Hills
Levittown
Tullytown
Bristol
Croydon
Eddystone
Crum Lynne
Ridley Park
Tinicum Township
Essington
Lester
Folsom
Darby Township
Collingdale
Clifton Heights
Aldan
Springfield
Cherry Hill
Golden Triangle
Erlton-Ellisburg
Greentree
Marlton
Moorestown
Cinnaminson Township
Riverton
Palmyra
Delanco
Riverside
Edgewater Park
Beverly
Burlington
Florence
Roebling
Mansfield
Delran
Riverton
Cinnaminson
Moorestown
Mount Laurel
Evesham
Marlton
Voorhees
Gibbsboro
Lindenwold
Stratford
Laurel Springs
Somerdale
Hi-Nella
Echelon
Ashland
Lawnside
Barrington
Bellmawr
Westville
Brooklawn
Mt Ephraim
Haddon Heights
Audubon Park
Audubon
Oaklyn
Collingswood
Woodlynne
Cherry Hill Mall
Merchantville
Pennsauken
Camden
Gloucester City
Brooklawn
Westville
Bellmawr
Barrington
Lawnside
Runnemede
Magnolia
Somerdale
Haddonfield
Haddon Heights
Mt Ephraim
Audubon
Collingswood
Woodlynne
Oaklyn
Barclay-Kingston
Erlton-Ellisburg
Golden Triangle
Cherry Hill
Cherry Hill Mall
Merchantville
Pennsauken
Moorestown-Lenola
Moorestown
Cinnaminson
Cinnaminson Township
Palmyra
Riverton
Riverside
Delanco
Edgewater Park
Bensalem
Feasterville-Trevose
Trevose
Oxford
Langhorne Manor
Langhorne
Bristol
Croydon
Tullytown
Fairless Hills
Levittown
Jenkintown
Wyncote
Rockledge
Willow Grove
Fort Washington
Fort Washington State Park
Flourtown
Wyndmoor
Oreland
North Hills
Glenside
Bryn Athyn
Plymouth Meeting
Plymouth Valley
Blue Bell
Ambler
Jamison
Upper Dublin
Lower Gwynedd
Spring House
Maple Glen
Horsham
Hatboro
Southampton
East Norriton
Norristown
West Norriton
Conshohocken
West Conshohocken
Bryn Mawr
Ardmore
Haverford
Havertown
Wynnewood
Penn Wynne
Narberth
Bala Cynwyd
Merion Station
Villanova
Radnor
Paoli
Malvern
Phoenixville
Valley Forge
Springfield
Yeadon
Lansdowne
Darby
Sharon Hill
Colwyn
Folcroft
Ridley Park
Prospect Park
Glenolden
Norwood
Morton
Rutledge
Media
Swarthmore
Wallingford
Newtown Square
Eddystone
Crum Lynne
Tinicum Township
Essington
Lester
Folsom
Darby Township
Collingdale
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